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Via EDIS

September 17, 2026

The Honorable Lisa R. Barton
Secretary to the Commission
United States International Trade Commission
500 E St., SW, Room 112
Washington, DC 20436

**Re: *Certain Semiconductor Devices, Products Containing Same, and Components Thereof,*
 Inv. No. 337-TA-1500**

Dear Secretary Barton:

Attached for filing on behalf of Respondents Tower Semiconductor Ltd.; Tower Partners Semiconductor Co., Ltd.; Tower Semiconductor Italy S.r.l.; Tower US Holdings, Inc.; Tower Semiconductor San Antonio, Inc.; Tower Semiconductor NPB Holdings, Inc.; Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC in the above-referenced investigation, please find a copy of Respondents' Markman Presentation.

Please contact me if you have any questions.

Respectfully Submitted,

/s/ Stephen R. Smith
Stephen R. Smith

Global Foundries v. Tower Semiconductor

Respondent's Presentation

September 17, 2026

Disputed Terms

1. “layer” / “first liner layer” / “refractory metal layer”
[’546 patent, claims 1–14; ’167 patent, claims 1, 3, & 4]
2. “intermix region”
[’546 patent, claims 1, 2, 6, 8, 9, & 13; ’167 patent, claims 1, 2, & 6]
3. “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide”
[’235 patent, claims 1 & 10]
4. “adjacent to”
[’983 patent, claims 1, 5, 13, 17, & 20]
5. “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate”
[’983 patent, claims 1, 13, & 20]
6. “the channel having a channel length that is controlled by ion implantation”
[’983 patent, claims 7 & 18]
7. “edge of the gate”
[’983 patent, claims 1, 2, 6, 13–15, & 20]

“layer” / “first liner layer” / “refractory metal layer”

'546 Patent, Claims 1-14

'167 Patent, Claims 1, 3, and 4

Disputed “Layer” Terms

Disputed Term

“layer” / “first liner layer” /
“refractory metal layer”

'546 patent, claims 1–14
'167 patent, claims 1, 3, and 4

Tower Semiconductor Proposed Construction

Indefinite or, if not indefinite:
“a separate and defined [first
liner / refractory metal] coating
of material”

GlobalFoundries Proposed Construction

Plain and ordinary meaning;
not indefinite.

Figure 6 from the Common Specification Depicts Exemplary Layers of Materials

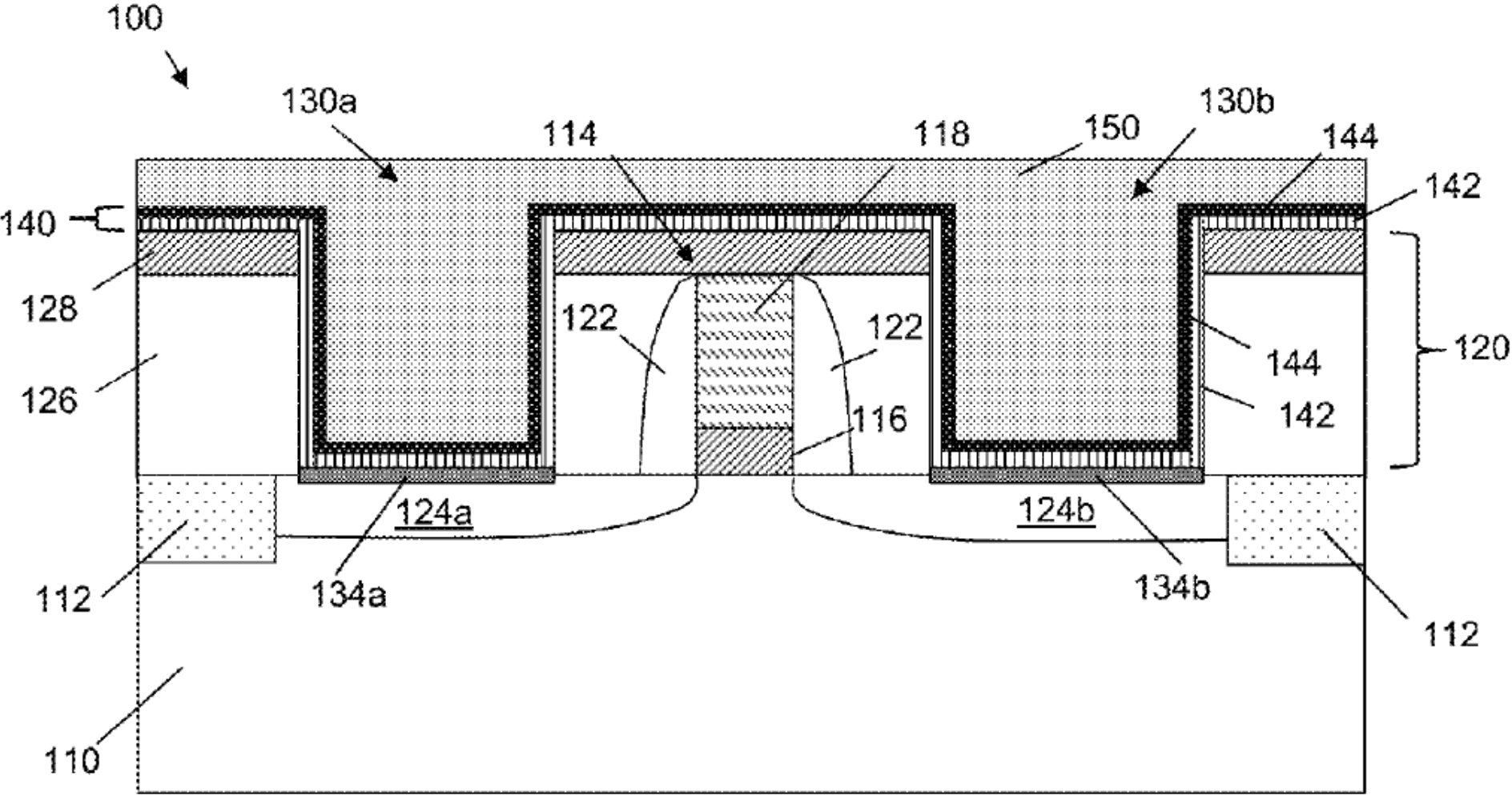


FIG. 6

Case Law: The CALJ Must Construe the Claims, Not Leave Disputes for Experts to Address

O2 Micro



“A determination that a claim term ‘needs no construction’ or has the ‘plain and ordinary meaning’ may be inadequate when a term has more than one ‘ordinary’ meaning or when reliance on a term’s ‘ordinary’ meaning does not resolve the parties’ dispute.”

O2 Micro Int’l Ltd. v. Beyond Innovation Tech. Co., 521 F.3d 1351, 1361 (Fed. Cir. 2008)

Case Law: The CALJ Must Construe the Claims, Not Leave Disputes for Experts to Address

Eon Corp.



“During trial, the parties’ experts disputed the meaning of the ‘portable’ and ‘mobile’ limitations. . . . Here, the court did not resolve the parties’ dispute by instructing the jury that the claims should be given their plain and ordinary meaning. During claim construction, the parties actively disputed the scope of the ‘portable’ and ‘mobile’ terms. . . . By determining only that the terms should be given their plain and ordinary meaning, the court left this question of claim scope unanswered This was legal error.”

Eon Corp. IP Holdings v. Silver Spring Networks, 815 F.3d 1314, 1317 (Fed. Cir. 2016)

The claims show that a “layer” is a separate and defined coating of material

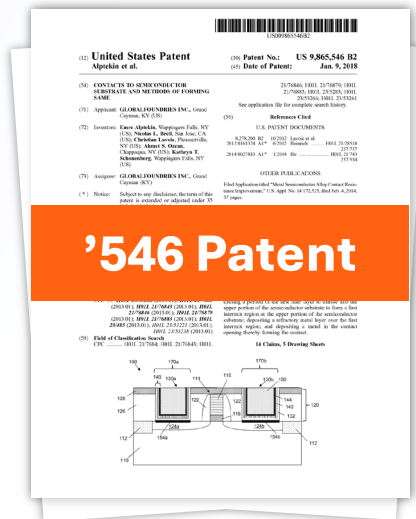
'546 Patent – Claim 1: All Layers Are Separate and Defined Coatings of Materials

1. A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;

depositing a first liner layer to conformally coat the contact opening causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;

depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.



'167 Patent – Claim 1: All Layers Are Separate and Defined Coatings of Materials

1. A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:

a contact opening in a dielectric layer over the semiconductor substrate;

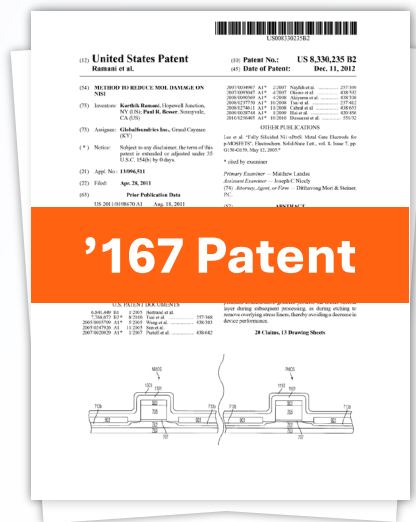
a first liner layer substantially coating a sidewall of the contact opening;

a refractory metal layer substantially coating the first liner layer in the contact opening;

a metal substantially filling the contact opening; and

an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate;

wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.



The specification confirms that a “layer” is a separate and defined coating of material

'546 Patent, Fig. 3 – The Dielectric and First Liner Layers Are Separate and Defined Coatings of Materials

1. A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate:

depositing a **first liner layer** to conformally coat the contact opening causing a portion of the **first liner layer** to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate:

depositing a refractory metal layer
over the first intermix region; and

depositing a metal in the contact opening thereby forming the contact.

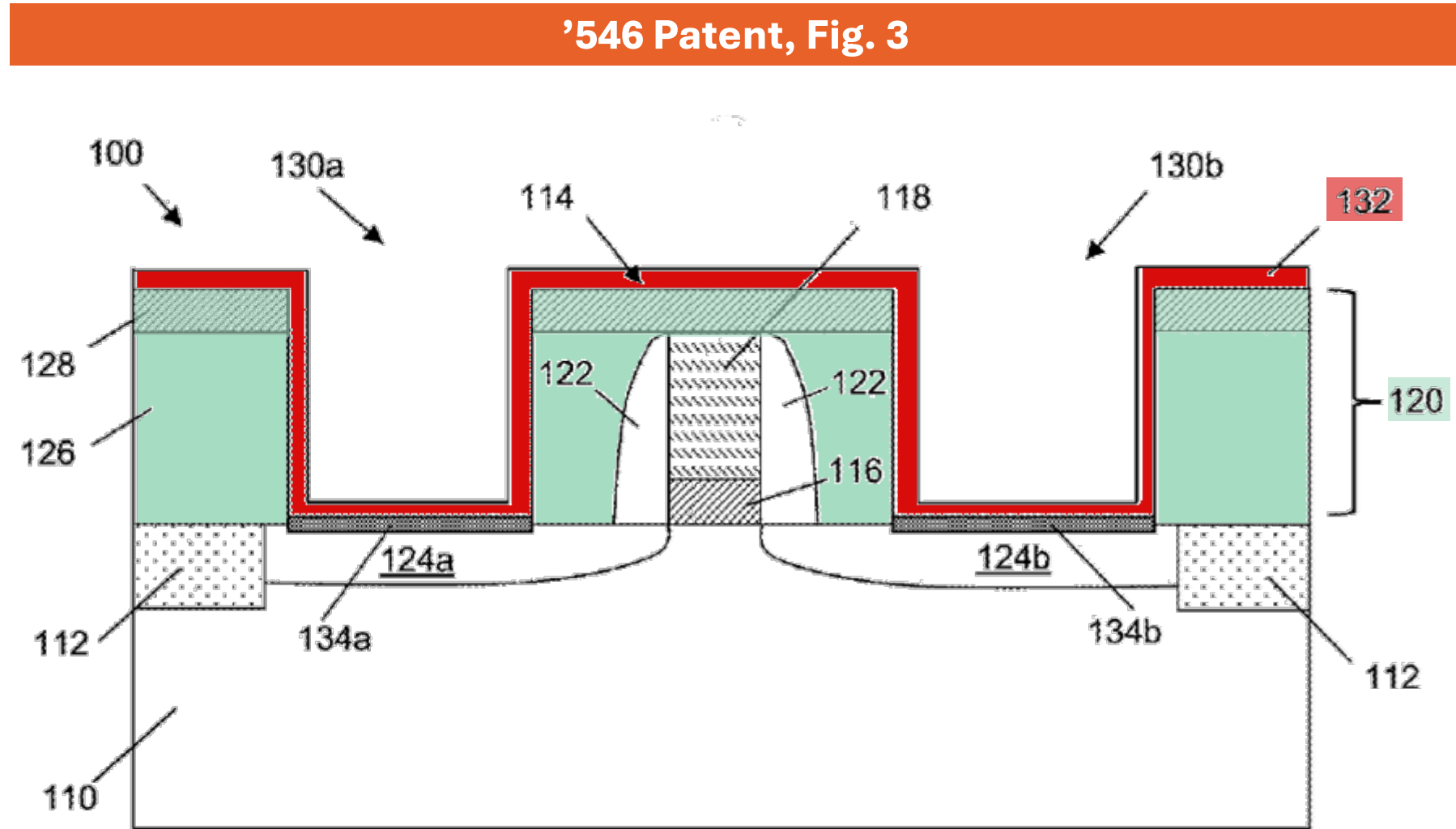


FIG. 3

“first liner layer 132” (’546 patent at 6:4–5)

“dielectric layer 120” (’546 patent at 4:57)

'546 Patent, Fig. 6 – The Refractory Metal Layer Is a Separate and Defined Coating of Material

1. A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate:

depositing a first liner layer to
conformally coat the contact opening

causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate:

depositing a refractory metal layer over the first intermix region; and

depositing a metal in the contact
opening thereby forming the contact.

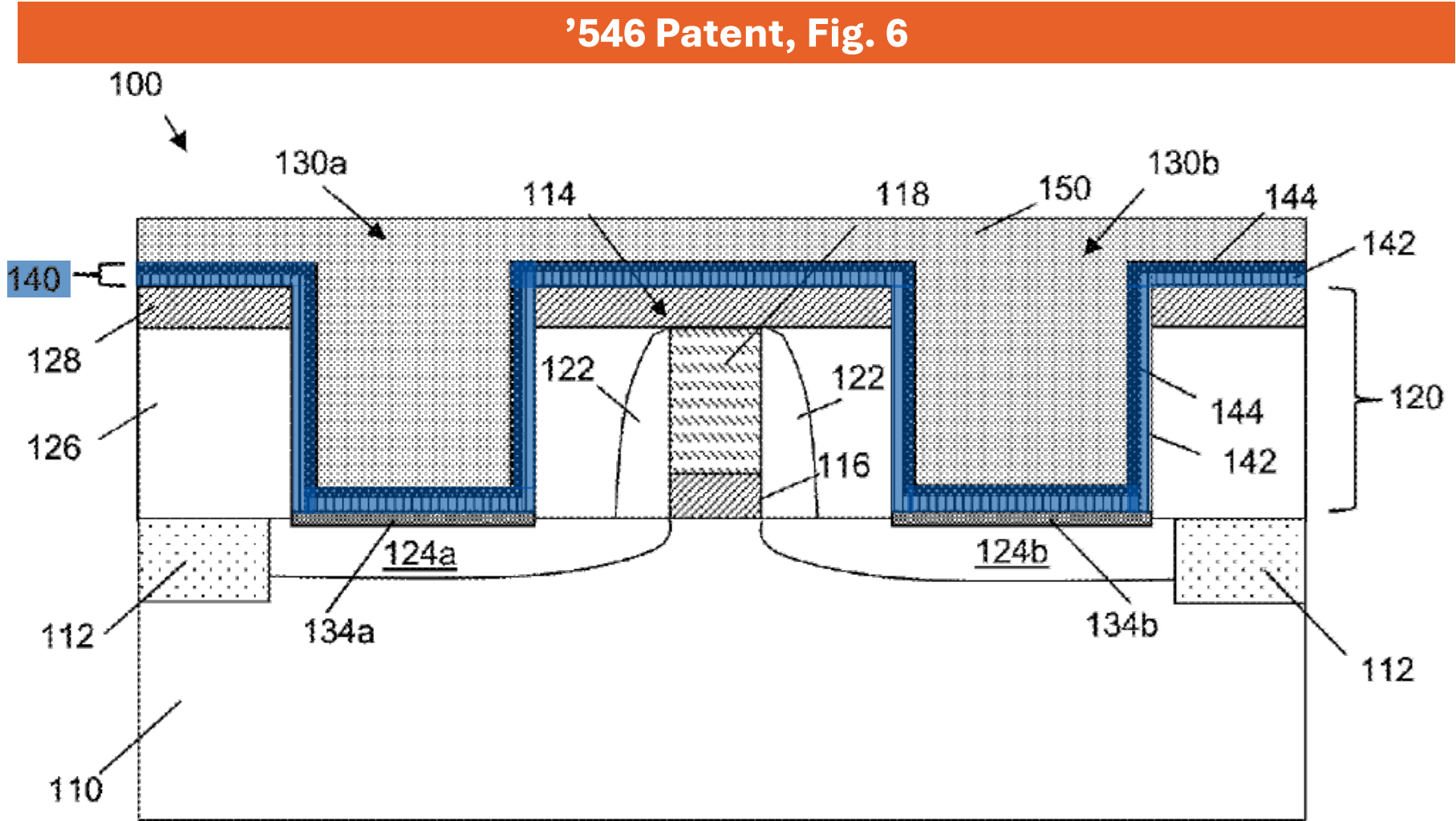


FIG. 6

“refractory metal layer 140” (’546 patent at 6:66)

'167 Patent, Fig. 3 – The '167 Patent's Specification Discloses the Same Embodiments of "Layers"

1. A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:
- a contact opening in a **dielectric layer** over the semiconductor substrate;
 - a **first liner layer** substantially coating a sidewall of the contact opening;
 - a **refractory metal layer** substantially coating the **first liner layer** in the contact opening;
 - a metal substantially filling the contact opening; and an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the **first liner layer**, a material used in the **refractory metal layer**, and a material used in the semiconductor substrate;
- wherein the **refractory metal layer** includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the **first liner layer** and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.

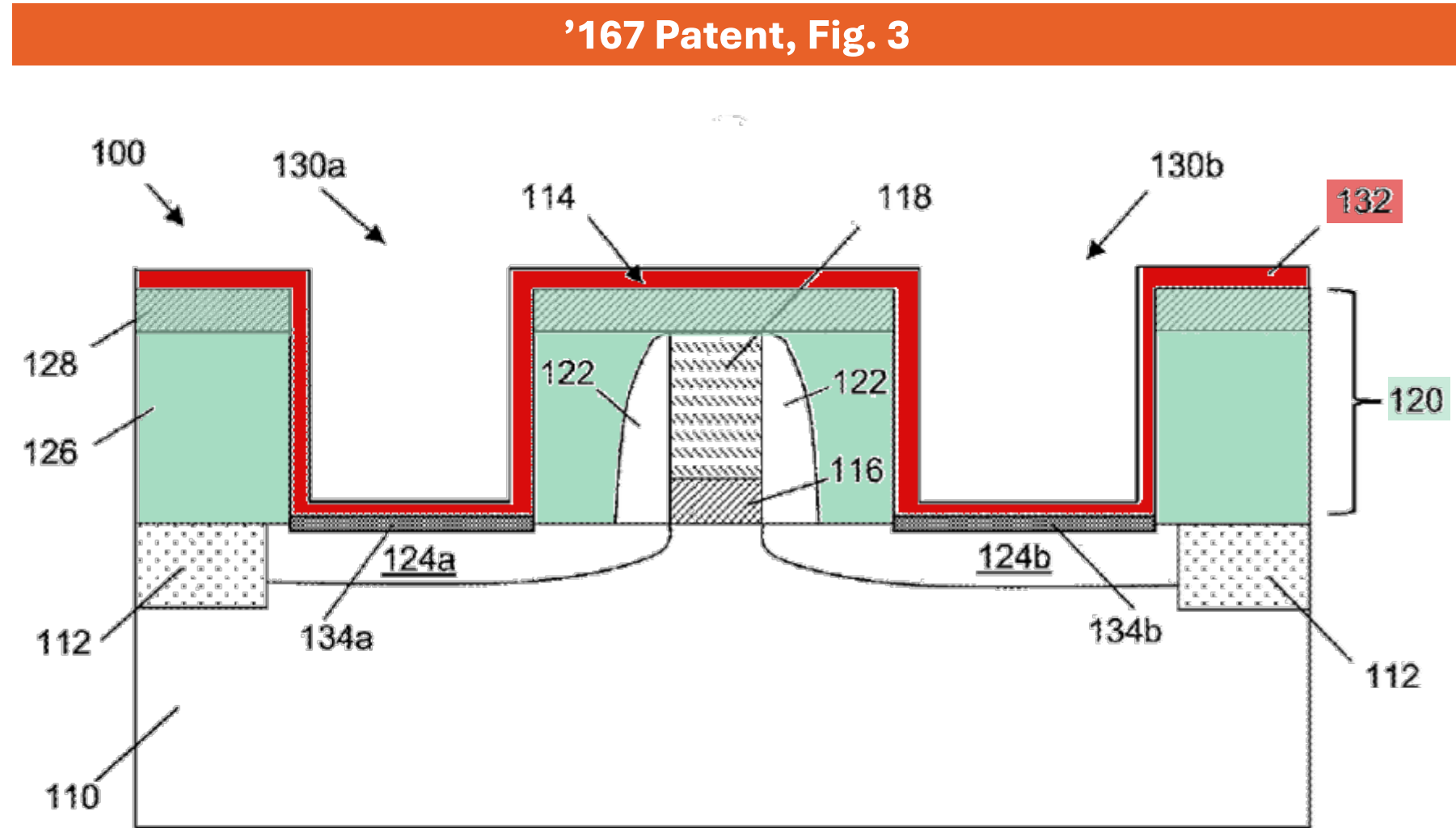


FIG. 3

“first liner layer 132” (’546 patent at 6:4–5)

“dielectric layer 120” (’546 patent at 4:57)

'167 Patent, Fig. 6 – The '167 Patent's Specification Discloses the Same Embodiments of “Layers”

1. A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:
- a contact opening in a **dielectric layer** over the semiconductor substrate;
 - a **first liner layer** substantially coating a sidewall of the contact opening;
 - a **refractory metal layer** substantially coating the **first liner layer** in the contact opening;
 - a metal substantially filling the contact opening; and an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the **first liner layer**, a material used in the **refractory metal layer**, and a material used in the semiconductor substrate;
- wherein the **refractory metal layer** includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the **first liner layer** and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.

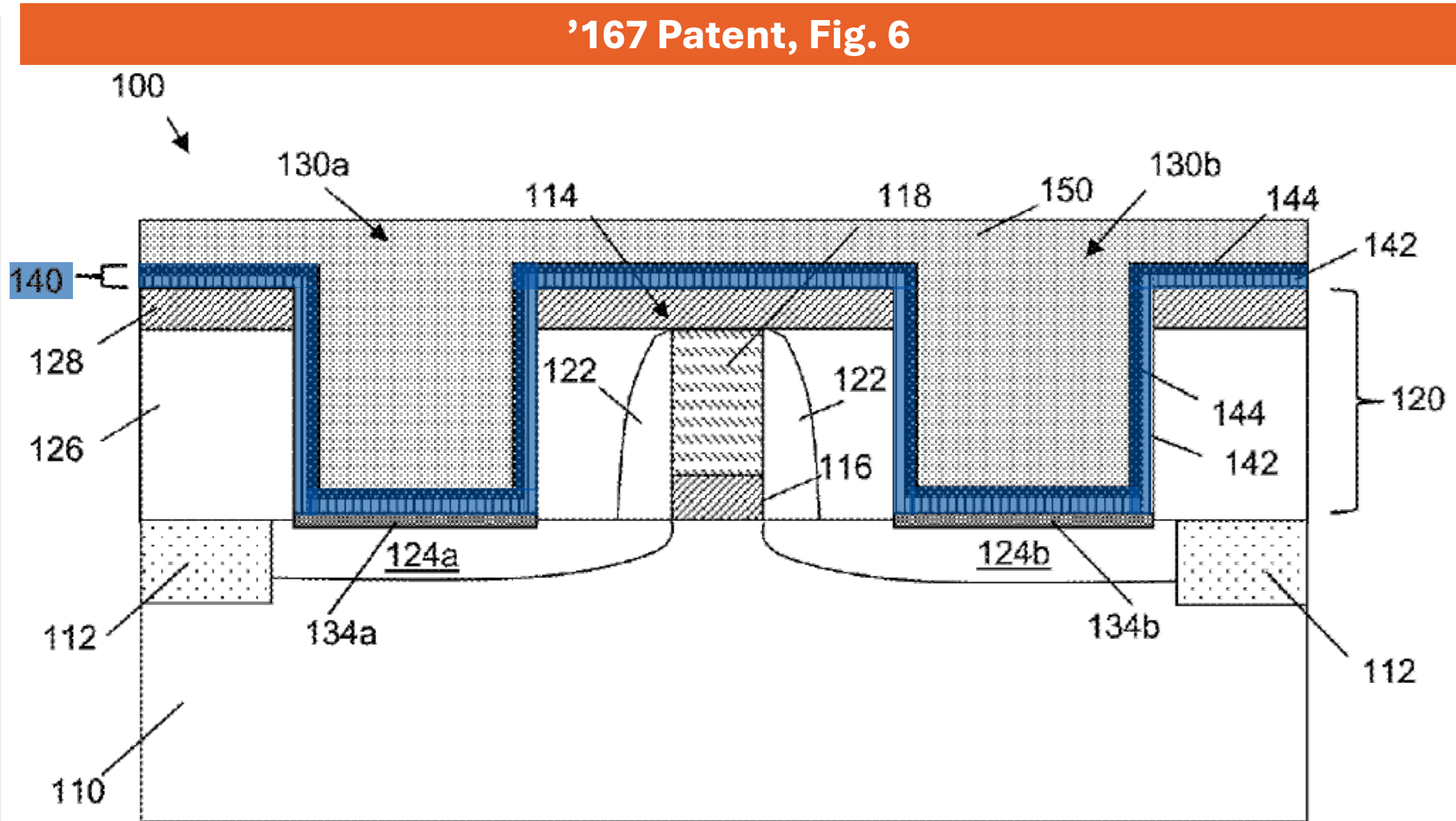


FIG. 6

“refractory metal layer 140” (’546 patent at 6:66)

Case Law: “Layers” Must Be “Separate and Distinct”

Advanced Tech. Incubator



“[I]f a ‘layer’ were not considered ‘**separate and distinct**,’ then the Parties could argue at trial that the term ‘layer’ be interpreted as several layers which are co-mingled and which are not considered either separate or distinct. Such a construction would contradict the ordinary meaning of the term ‘layer.’”

Advanced Tech. Incubator, Inc. v. Sharp Corp., C.A. No. 2:7-cv-00468, Claim Constr. Ord. at 17 (E.D. Tex. Mar. 11, 2009), Dkt. No. 117

Case Law: “Layers” Are Defined Coatings of Material

Intellectual Ventures



“Webster’s dictionary defines ‘layer’ as ‘[a] single thickness, **coating**, or stratum spread out or covering a surface.’”

Intellectual Ventures I LLC, v. Canon Inc., C.A. No. 11-792-SLR, Memo. Ord. re Claim Constr. at 9–10 (D. Del. Apr. 10, 2014), Dkt. No. 252 (construing “layer” to mean “[a] **coating of material** deposited or formed across a surface”)

Bridgelux



“Wiley defines ‘layer’ as ‘[a] **defined** thickness which is part of a **material**, or which surrounds it,’ citing as an example, ‘a layer in a semiconductor.’ Wiley at 413. This basic definition is consistent with the hundreds of individual uses of ‘layer’ in the patents-in-suit.”

Bridgelux, Inc., v. Cree, Inc., C.A. No. 06-6495-PJH, Ord. Construing Claims at 10 (N.D. Cal. Aug. 15, 2008), Dkt. No. 109

Respondents' Construction Does Not Exclude Embodiments from the Specification

Complainant argues that Respondents' construction reads out certain embodiments:

1. A dielectric layer that includes multiple sub-layers of dielectric materials
2. A refractory metal layer that includes multiple sub-layers of refractory metal materials
3. A first liner layer that diffuses into the substrate to form an intermix region

Respondents' Construction Covers Dielectric Bilayers

- Complainant argues that Respondents' construction excludes dielectric bilayers
- Respondents' construction covers dielectric bilayers 128 and 126 that are separate and defined, as described in the specification

Still referring to FIG. 1, a dielectric layer 120 may be formed such that dielectric layer 120 substantially surrounds gate 114. In one embodiment of the invention, formation of dielectric layer 120 may include deposition of a dielectric material 126 and a dielectric cap material 128. While shown as two separate layers, it is to be understood that dielectric layer 120 is not limited to only two layers. Rather, in other embodiments of the invention, dielectric layer 120 may include a single layer or more than two separate layers.

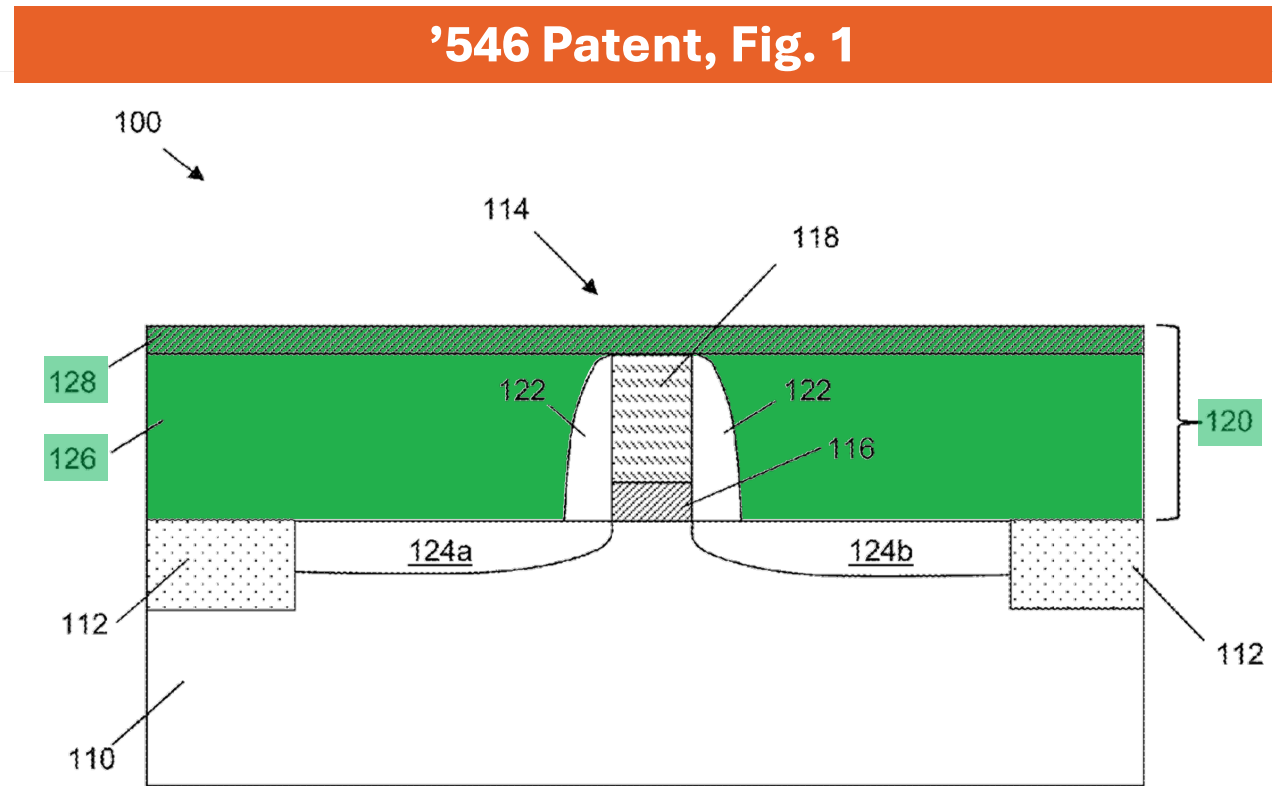


FIG. 1

Respondents' Construction Covers Refractory Metal Bilayers

- Complainant argues that Respondents' construction excludes refractory metal bilayers
- Respondents' construction covers refractory metal bilayers 142 and 144 that are separate and defined, as described in the specification

In some embodiments, as shown in FIG. 5, **refractory metal layer 140** may include a bilayer of a **first refractory material 142** and a **second refractory material 144** such that first refractory metal material 142 substantially separates first liner layer 132 from second refractory metal layer 144. First and second refractory metal materials 142, 144 may include any of the refractory metal materials previously mentioned. For example, in some embodiments, first refractory metal material 142 may include Ti and second refractory metal material 144 may include TiN.

'546 Patent, Fig. 5

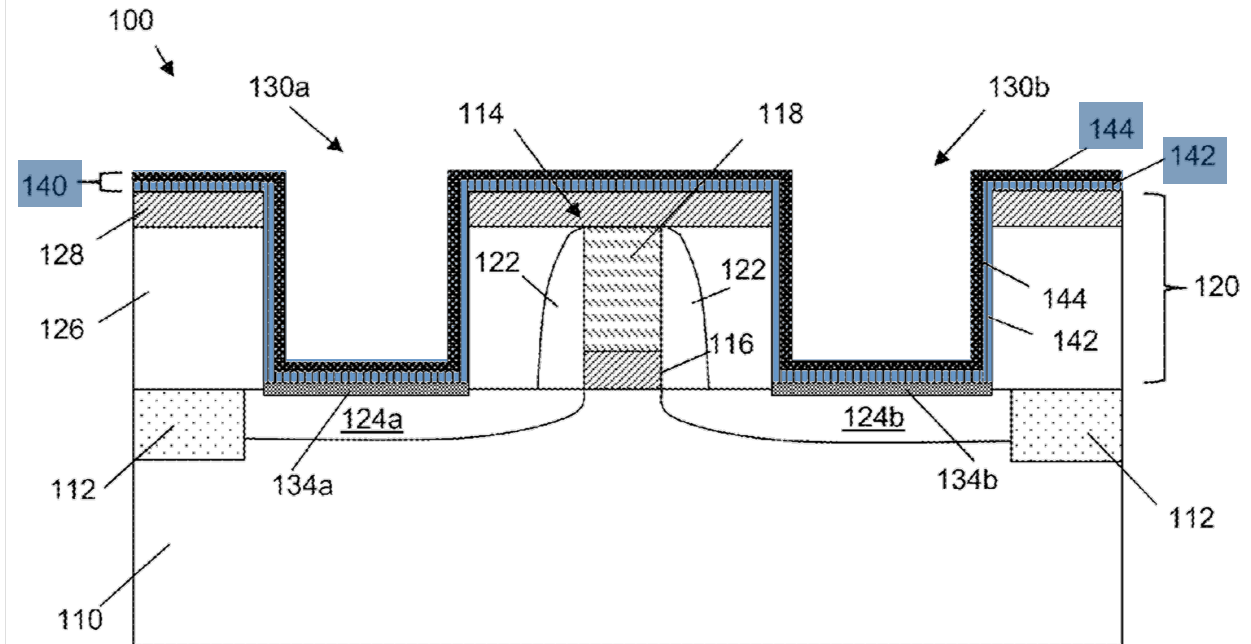


FIG. 5

Respondents' Construction Covers First Liner Layers

- Complainant argues that Respondents' construction excludes a first liner layer that diffuses into the substrate to form an intermix region.
- Respondents' construction covers the undiffused portion of the first liner layer 132 being a "first liner layer," as described in the specification. The "intermix region" is a separate element.

Referring now to FIG. 4, another step of the method according to embodiments of the invention is shown. In this step, **first liner layer 132** is removed from each contact opening 130a, 130b as well as from the upper surface of dielectric cap material 128. That is, the remaining portion of **first liner layer 132** which does not diffuse into the respective source 124a and drain 124b is removed.

'546 Patent, Fig. 3

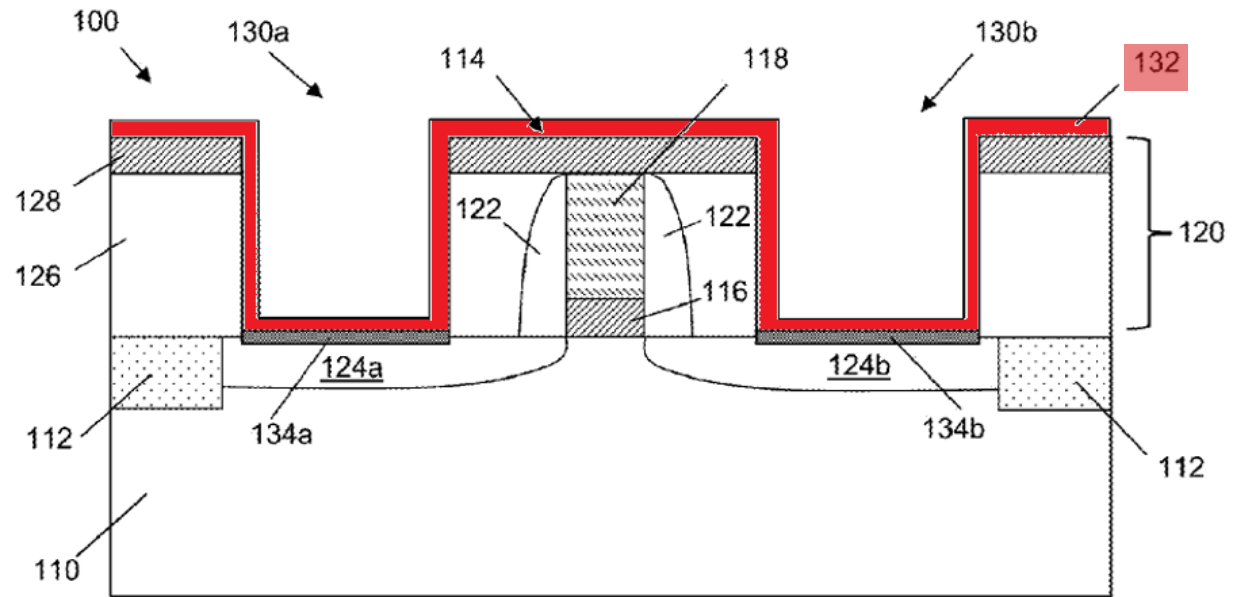
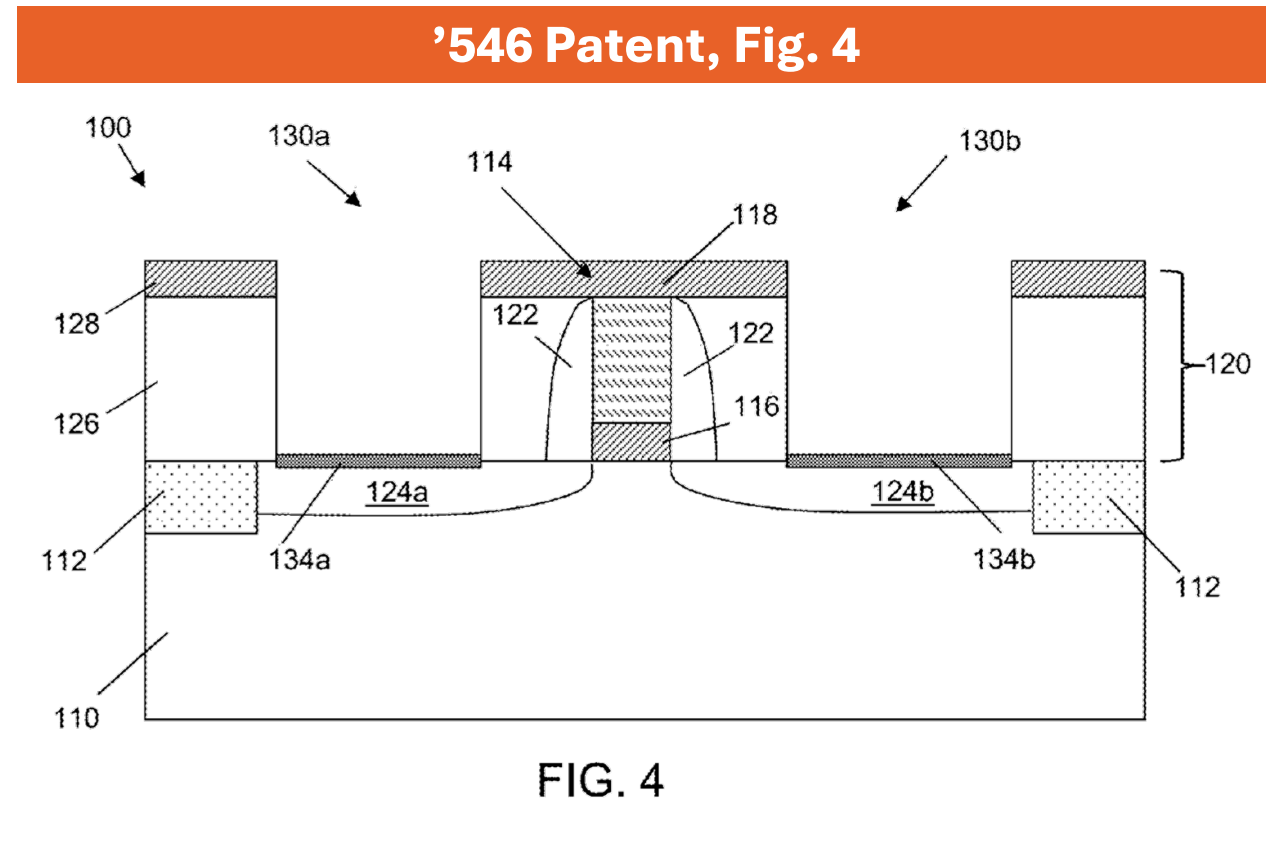


FIG. 3

Respondents' Construction Covers First Liner Layers

- Complainant argues that Respondents' construction excludes a first liner layer that diffuses into the substrate to form an intermix region.
- Respondents' construction covers the undiffused portion of the first liner layer 132 being a "first liner layer," as described in the specification. The "intermix region" is a separate element.

Referring now to FIG. 4, another step of the method according to embodiments of the invention is shown. In this step, **first liner layer 132** is removed from each contact opening 130a, 130b as well as from the upper surface of dielectric cap material 128. That is, **the remaining portion of first liner layer 132** which does not diffuse into the respective source 124a and drain 124b is removed.



Respondents' Construction Covers the Disclosed Bilayer Embodiments

I disagree with Dr. Liu's statement that Respondents' alternative proposed construction would exclude any embodiment in the specification of the '546 or '167 patents. **Respondents' alternative proposed construction does not require a coating of material that is uniformly made up of only a single element or alloy.** Liu Decl., ¶ 38. Instead, the material could comprise a bilayer of Ti/TiN for refractory metal purposes or a bilayer of different dielectric materials, so long as the larger "layer" is identifiable by its purpose such that it is separate and defined. See '546 patent, 4:54-5:23 (describing potential bilayer of different dielectric materials, which the POSA would collectively recognize as being a "dielectric layer" because of the collective purpose of such layers to serve as an electrically insulating material whose main purpose is to control electric fields while preventing unwanted current flow); id., 6:64-7:23 (describing a bilayer of Ti/TiN that would have been known to serve adhesion (Ti) and barrier (TiN) purposes in the contact stack). **In other words, the "material" referenced in Respondents' alternative proposed construction need not be a uniform thickness of a single element or alloy. It could be made up of different elements or alloys, so long as the identifiable "material" of the layer is separate and defined, such that a POSA could distinguish it from other layers or components.** Dr. Liu does not explain how a POSA could understand the scope of the claims with reasonable certainty if the claimed layers are not separate and defined.

Respondents' Construction Covers the Disclosed Embodiments of First Liner Layers

Dr. Liu asserts that Respondents' alternative proposed construction excludes the embodiment of a first liner layer that partially diffuses into an intermix region. Liu Decl., ¶ 38. I disagree. The first liner layer is a separate and defined coating of material in both the '546 and '167 patents' claims.

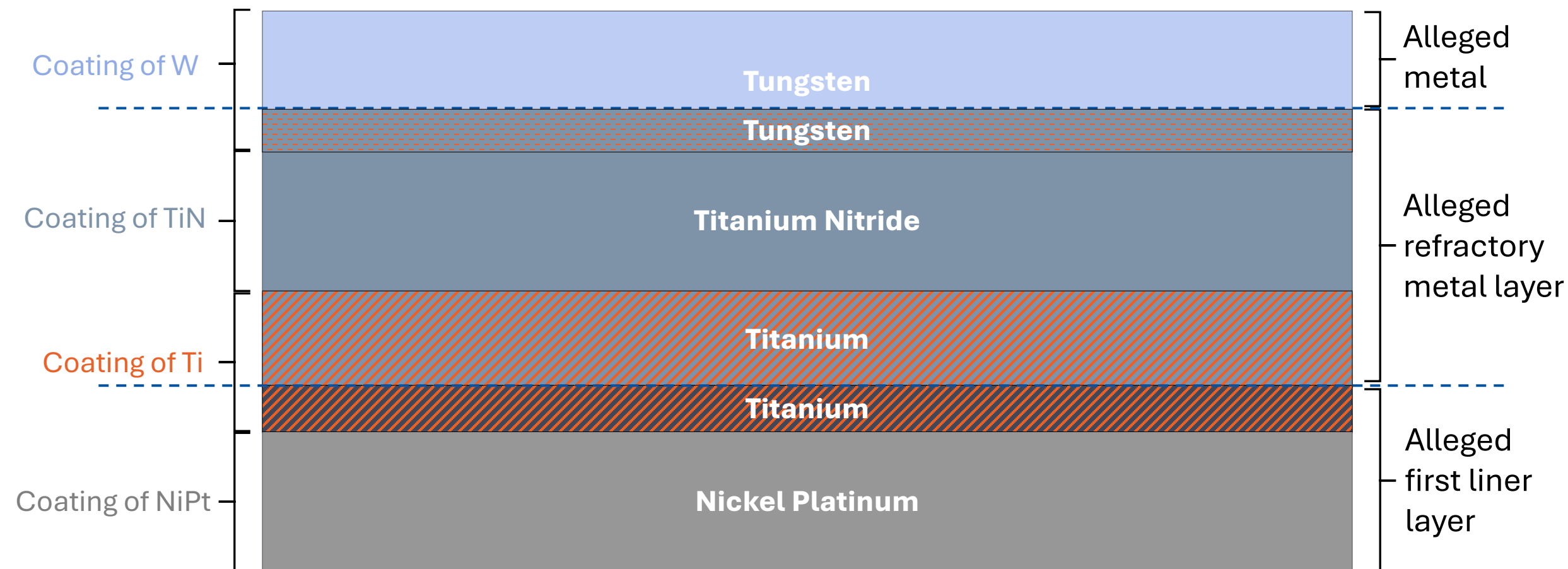
The '167 patent requires an intermix region “having a composition including a material used in the first liner layer.” '167 patent, claim 1. This does not mean that the first liner layer itself, as claimed, is inside the intermix region. A POSA would have understood the term “intermix region” to describe a separate component of the claimed semiconductor structure that includes *material used* in the first liner layer, not the first liner layer itself (which is recited as being a separate and defined coating of material).

Likewise, the '546 patent's claims recite method steps, one of which recites either “depositing a first liner layer to conformally coat the contact opening” in claim 1 and its dependent claims or “depositing a first liner layer to conformally coat each contact opening” in claim 8 and its dependent claims. A POSA would understand this separate and defined coating of material to be a first liner layer.

The fact that a subsequent method step recites creation of a “first intermix region” that includes a portion of the first liner layer does not change the fact that a POSA would have understood the first liner layer, as originally deposited, to be a separate and defined coating of material.

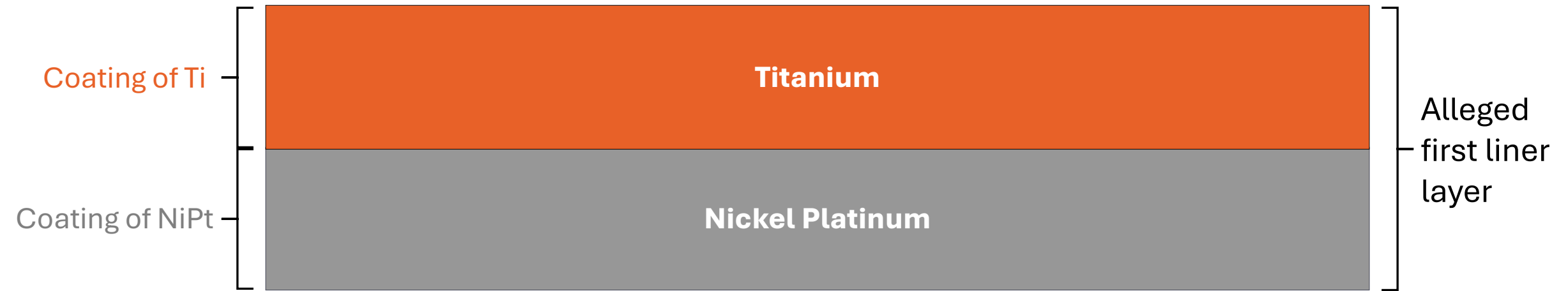
Complainant's Plain and Ordinary Construction Is Indefinite

Complainant's Tech DI Contentions Arbitrarily Divide and Combine Alleged “layers”



A POSA could not identify these arbitrarily designated “layers” with reasonable certainty because they are not separate and defined

Complainant's Infringement Contentions Arbitrarily Combine Alleged “layers”



A POSA could not identify these arbitrarily designated “layers” with reasonable certainty because they are not separate and defined

“intermix region”

'546 Patent, Claims 1, 2, 6, 9, and 13

'167 Patent, Claims 1, 2, and 6

Disputed “Intermix Region” Term

Disputed Term

“intermix region”

'546 Patent, Claims 1, 2, 6, 9, and 13
'167 Patent, Claims 1, 2, and 6

Tower Semiconductor Proposed Construction

Indefinite

GlobalFoundries U.S. Proposed Construction

Plain and ordinary meaning;
not indefinite.

Patents Are Required to Provide Objective Boundaries

Inv. No. 337-TA-1397

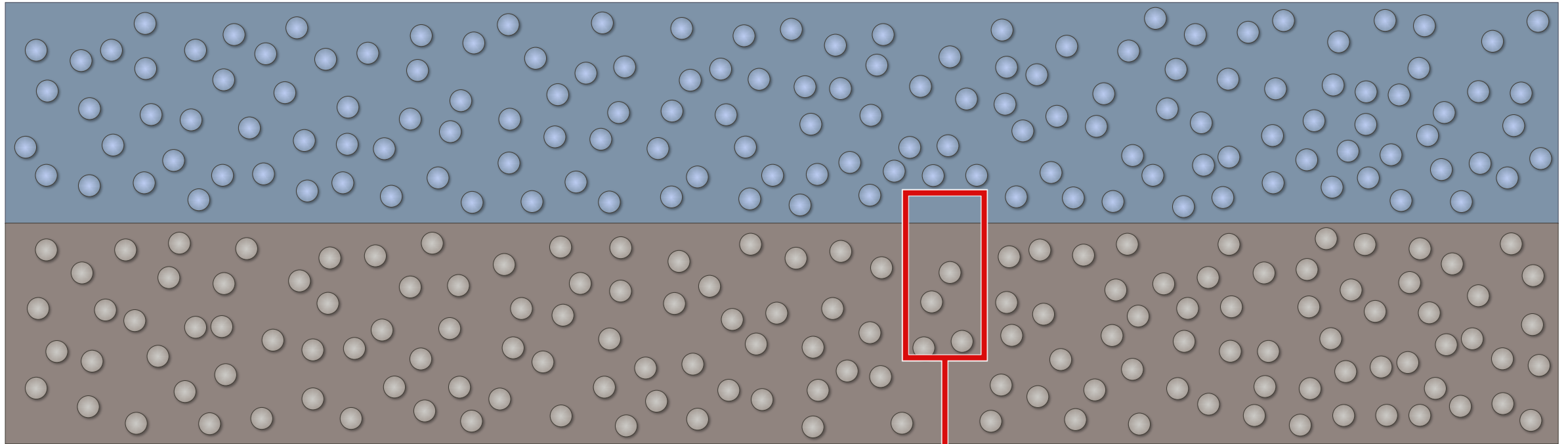


“A claim is required to provide objective boundaries for those of skill in the art, and a claim term may be indefinite if it might mean several different things and no informed and confident choice is among the contending definitions.”

*Certain Cellular Base Station Commc'n Equip., Components Thereof, & Prods. Containing Same, Inv. No. 337-TA-1397, Ord. No. 16, 2024 WL 5245312, at *4 (Nov. 5, 2024) (quoting Interval Licensing LLC v. AOL, Inc., 766 F.3d 1364, 1371 (Fed. Cir. 2014) (citation modified)).*

How much intermixing is required to form an “intermix region”?

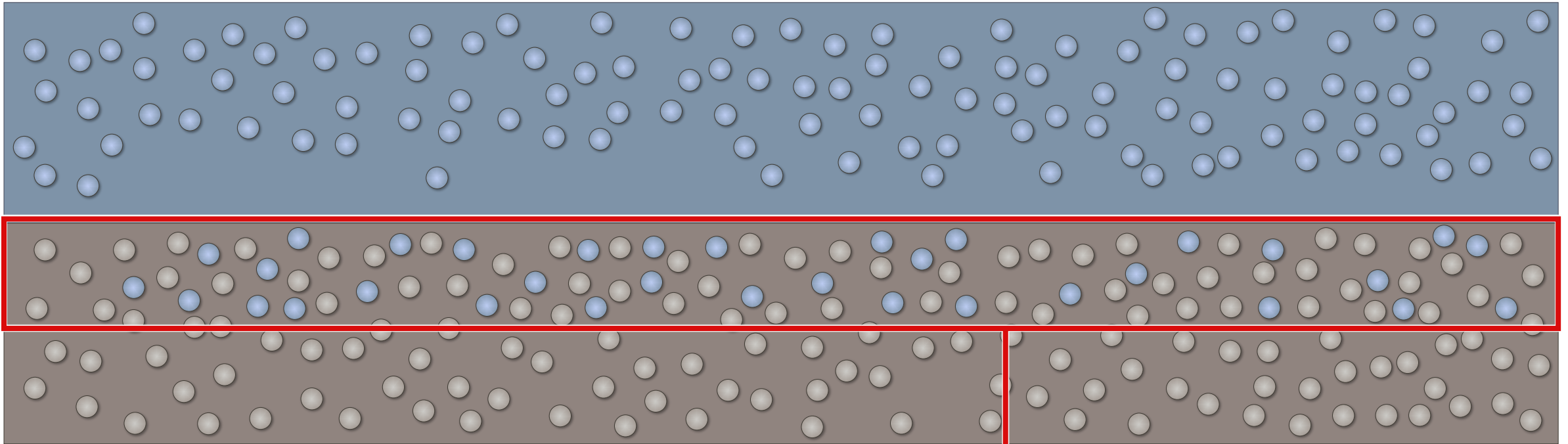
A: First liner layer



B: Semiconductor substrate

Intermix region?

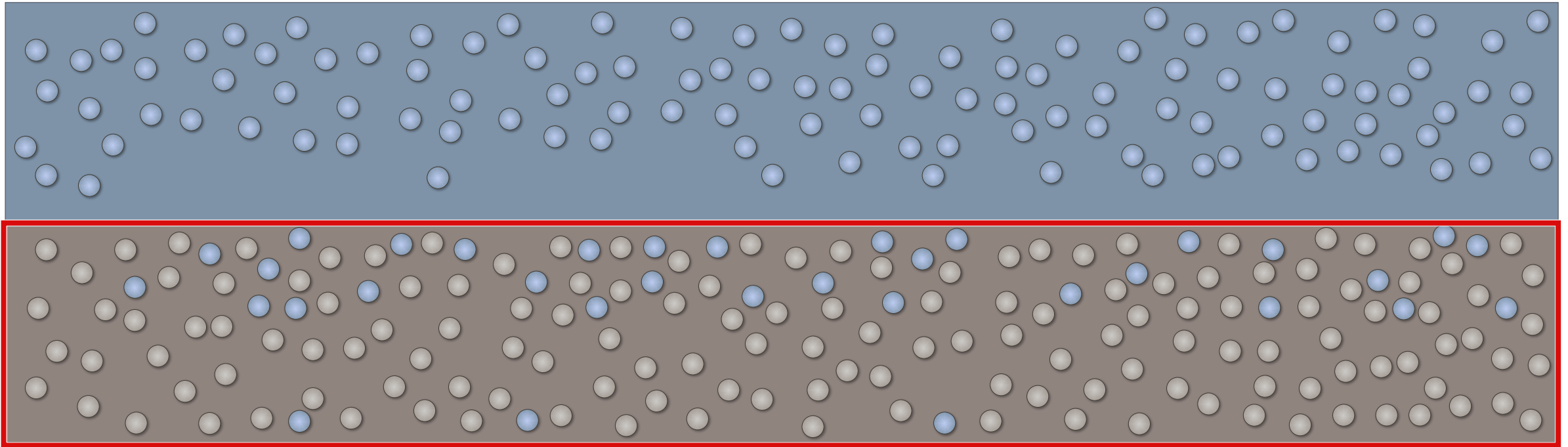
A: First liner layer



B: Semiconductor substrate

Intermix region?

A: First liner layer

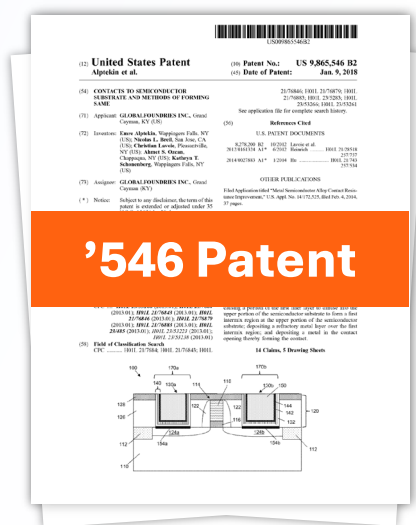


B: Semiconductor substrate

Intermix region?

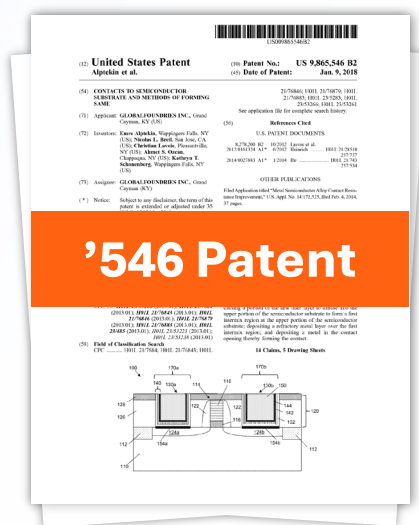
The Common Specification Lacks Objective Boundaries

In one example, contacts **170a** and **170b** may include NiPt as first liner layer **132**, a Ti/TiN bilayer as refractory metal layer **140**, and W as metal **150**. In this embodiment, first intermix regions **134a**, **134b** may include a NiPt silicide and second intermix regions **154a**, **154b** may include a NiPtTi silicide. However, it is to be understood that other combinations of materials may be used to create contacts **170a**, **170b** without departing from aspects of the invention. For example, in another embodiment, first liner layer **132** may include Ni, refractory metal layer **140** may include a Ti/TiN bilayer, and metal **150** may include W. In this embodiment, first intermix regions **134a**, **134b** may include a Ni silicide and second intermix regions **154a**, **154b** may include a NiTi silicide.

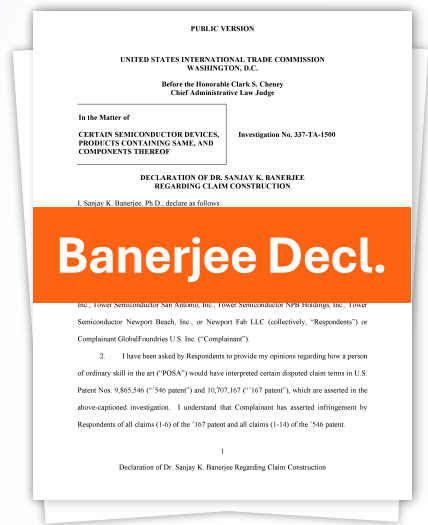


The Common Specification Lacks Objective Boundaries

First intermix regions **134a**, **134b** may originate from the high energy of the incoming sputtered atoms (from first liner layer **132**) which are thus pushed into the Si lattice of source **124a** and drain **124b**. Therefore, first intermix regions **134a**, **134b** may include a mixture of the material used for first liner layer **132** and a material used for the respective semiconductor substrate **110** thereunder. In the embodiment shown in FIG. 3, first intermix regions **134a**, **134b** may have a composition including a material used for first liner layer **132** and materials used for the respective source and drain **124a**, **124b** thereunder. While the Si lattice is a dense way to pack Si atoms, it has a relatively open structure due to its bonding being covalent. The first few metal atoms, arriving with relatively high energy, can be forced into the interstitial sites of the Si lattice to create a thin region (intermix regions **134a**, **134b**) of mixed Si and metal on the order of 3-5 nm in thickness, for example. In some cases, first intermix region **134a**, **134b** may have a relatively high Si content (greater than 50% Si in atomic composition). In other



Expert Evidence Confirms the Patents Lack Objective Boundaries



80. Even to the extent the patents provide guidance on what it means to be “intermixed” (they do not), the patents provide no guidance on the *amount* of “intermixing” that is required. For example, assuming that “intermixing” includes diffusion of atoms from one coating into another coating, is there some threshold level of diffusion that is required? The patents include a statement that “very little if any metal diffusion is needed” to form the intermix region, but lack any guidance on what qualifies as “very little.” ’546 patent, 6:39-43. For example, if a single atom from one layer diffuses into another layer, is this sufficient to qualify as an “intermix region”? If a region is 98% of one material, and only 2% of a second material, is this enough to comprise an “intermix region”? One portion of the common specification notes that the “intermix region” may have around 50% mixing between two materials. *Id.*, 6:19-21. But the specification goes on to identify numerous factors that may impact the amount of intermixing. *Id.*, 6:21-28 (“type of material being deposited, the composition of source 124a and drain 124b, the energy of the atoms of the material being deposited, and/or the conditions of the deposition technique (temperature, pressure, reaction, atmosphere, etc.)”), 6:28-31 (“etch selectivity”), 6:31-34 (“amorphous” versus “crystalline”). While the common specification identifies factors that may impact the amount of intermixing, it puts no meaningful limits on what amount of mixing would or would not qualify as creating an “intermix region.”

Expert Evidence Confirms the Patents Lack Objective Boundaries

Dr. Liu

GlobalFoundries
Technical Expert

Q. How much intermixing is required in order for it to qualify as an intermix region?

...

A. ...

The problem does provide certain context that to the extent the intermix region—I think your question is how much intermixing. It really is—is—is context-dependent, within the context of forming a liner layer inside a contact opening.

Expert Evidence Confirms the Patents Lack Objective Boundaries

Dr. Liu

GlobalFoundries
Technical Expert

Q. So you provided a couple of examples. You talked about the thickness, and you said 3 to 5 nanometers thick.

If it's only 2 nanometers thick, does it not qualify as an intermix region?

...

A. Look, I give an example of 3 to 5 nanometers based on what the patent specification cited. That gives me a clear understanding to determine the scope and meaning of the term "intermix region."

Q. And if it's only 2 nanometers, does it fall outside of the claim, then?

A. I will have to analyze it. ... I haven't done analysis. I haven't done analysis for 2 nanometers, necessarily, or 1 nanometer. I have to see the context of the situation, the context of the -- of -- I have to have the full context for me to do analysis.

Expert Evidence Confirms the Patents Lack Objective Boundaries

Dr. Liu

GlobalFoundries
Technical Expert

Q. In your response, you also referenced this—the composition of the intermix region and the discussion in the specification of a high silicon content, greater than 50 percent of silicon.

If it was 98 percent silicon, would it still be an intermix region?

. . .

A. Look, again, this is what the patent specification cited to help a POSITA to grasp and determine the scope and meaning of the term “intermix region,” and it helps to determine that this term is not indefinite, with reasonable certainty, to—for the scope and meaning.

So you want to throw out a hypothetical, 90 percent. I have to look at the context. I have to look at exactly—again, within the '546 patent, of forming a liner layer on top of the contact opening. I have to see the context. I have to see—I have to do the analysis. I have to see the context in order to answer your question.

“from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide”

'235 Patent, Claims 1 and 10

Disputed Terms

Disputed Term

“from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide”

'235 Patent, Claims 1 & 10

Tower Semiconductor Proposed Construction

Indefinite

GlobalFoundries U.S. Proposed Construction

Plain and ordinary meaning;
not indefinite

Different Methodologies, Yielding Different Results, Renders Claims Indefinite

Dow Chem. Co. v. Nova Chems. Corp.,



“The question is whether **the existence of multiple methods leading to different results** without guidance in the patent or the prosecution history as to which method should be used **renders the claims indefinite**. . . . Here the required guidance is not provided by the claims, specification, and prosecution history.”

Dow Chem. Co. v. Nova Chems. Corp., 803 F.3d 620, 634-35 (Fed. Cir. 2015) (emphasis added).

Different Methodologies, Yielding Different Results, Renders Claims Indefinite

Teva Pharms. USA, Inc. v. Sandoz, Inc.



“To summarize, it is undisputed that ‘molecular weight’ or average molecular weight can be ascertained by any of three possible measures: M_p , M_n , and M_w . The claims do not indicate which measure to use. The specification never defines molecular weight or even mentions M_p , M_w , or M_n We hold that claim 1 is invalid for indefiniteness by clear and convincing evidence because read in light of the specification and the prosecution history, the patentee has failed to inform with *reasonable certainty* those skilled in the art about the scope of the invention. On this record, there is not reasonable certainty that molecular weight should be measured using M_p . This is the legal question — and on this question — we reverse the district court.”

Teva Pharms. USA, Inc. v. Sandoz, Inc., 789 F.3d 1335, 1343-45 (Fed. Cir. 2015).

Complainant's Case Law Is Inapt

Ironburg Inventions Ltd. v. Valve Corp.



“[W]hat rendered the claims in *Dow* and *Teva* indefinite was that ‘the challenger ha[d] shown that there were competing existing methodologies that reached different results, and the patent failed to describe which of the multiple methods to use.’ **Here, by contrast, both parties’ experts agreed on and used the same measurement methodology** – which was simply to find the distance between two points. That is, **unlike the parties in *Dow* and *Teva***, the disagreement between Ironburg and Valve is about the application of **an agreed-upon, straightforward methodology.**”

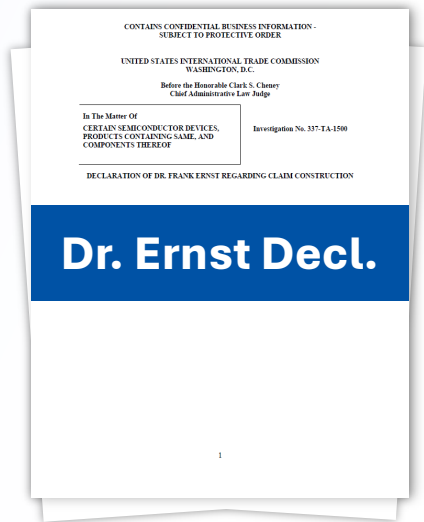
Ironburg Inventions Ltd. v. Valve Corp., 64 F.4th 1274, 1289—90 (Fed. Cir. 2023) (emphasis added).

Undisputed Evidence – Different Methodologies, Yielding Different Results

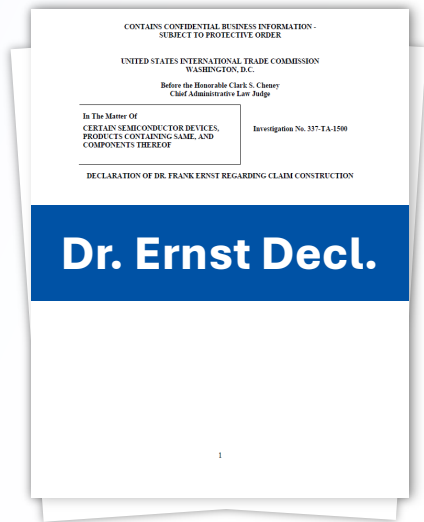
37. In my opinion, a POSA would not have been reasonably certain as to where the “nickel silicide surface at a silicon interface” is located based on the language of the specification.

The prosecution history also fails to provide any guidance. In particular, the '235 patent does not provide any guidance as to how the location of the “nickel silicide surface at a silicon interface” is to be *measured*.

38. A number of acceptable methods of “microcharacterization” are in use for this purpose, but they yield *different* results. For example, researchers have used XEDS (X-ray Energy-Dispersive Spectrometry) in conjunction with SEM (Scanning Electron Microscopy) to measure profiles $X_i[z]$ or $W_i[z]$ of the atom fraction X or the mass fraction W of elements $i, i = 1 \dots N$, and determine surface/interface positions from the characteristics of the profiles.



Undisputed Evidence – Different Methodologies



2. Methodologies for Locating an Interface

52. A POSA would understand there are multiple methodologies for locating an interface.

53. For example, a paper entitled “Silicide-phase evolution and platinum redistribution during silicidation of $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ specimens” by Praneet Adusumilli, David N. Seidman, and Conal E. Murray (“Adusumilli”) (Ex. 17) explores the Si/silicide interface using Atomic Probe Tomography (APT). Ex. 17 at 1. Using APT, the authors obtained highly accurate topographical 3D maps of *arbitrarily* chosen Si reference levels, e.g. $X_{\text{Si}} = 0.75$, a “75 at. % Si

• • •

56. A different approach for locating the position of the “silicide surface” and the “silicon interface” emerges from the ability of TEM (transmission electron microscopy) not only to produce images and spectra, but also electron diffraction patterns from selected areas. These electron diffraction patterns form in the back-focal plane of the first lens after the specimen, called the objective lens. To switch the microscope from imaging the image plane of the objective lens to imaging the back focal plane of the objective lens onto the viewing screen, all that is needed is to adjust the focal length of the lens below, which is called the intermediate lens. Every transmission microscope can do this switch by the push of a button. If the selected area from

Undisputed Evidence – Different Methodologies

Dr. Ernst

Tower Semiconductor
Technical Expert

- Q. But here is my question: Dr. Ernst, for the purposes of your claim construction opinions, what are the various techniques you identify for determining where a -- where the silicon interface is located?
- A. Okay. So to start this, there are various different techniques of microcharacterization as I call them. There are techniques of imaging, there are techniques of spectrometry, and there are techniques of diffractometry. And in this document, the three different parts you looked at or that you just recited, they show some examples without going into details what some of these techniques are in the first part. ...
And I want to say that the first overview is not even exhaustive. There are many more techniques.

Undisputed Evidence – Different Methodologies

Dr. Ernst

Tower Semiconductor
Technical Expert

- Q. Dr. Ernst, do you recall being asked about a number of different methodologies that a person of ordinary skill in the art could have used to determine the location of the silicon interface?
- A. Yes. I recall that, and I can extend the list I gave earlier if you want me to.

Undisputed Evidence – Different Methodologies

Dr. Ernst

Tower Semiconductor
Technical Expert

Q. Well, do you recall being asked about SEM XEDS?

A. Yes.

...

Q. And do you recall testifying with respect to a methodology that you referred to as SEM XWDS?

A. Yes, I do.

...

Q. And do you recall providing testimony regarding a methodology that you referred to as Rutherford backscattering?

A. Yes, I do.

...

Q. And do you recall providing testimony regarding x-ray photoelectron spectrometry which you've shortened to XPS?

A. Yes, I do.

...

Q. And do you recall being asked about transmission electron microscopy?

A. I do.

Q. And do you recall providing some testimony about a methodology you referred to as ToF-SIMS?

A. Yes, I do.

...

Q. And do you recall providing some testimony on what you referred to as Auger profiling?

A. Yes. Scanning Auger Microprobe which is basically an SEM that scans a focused electron beam over the specimen surface in ultrahigh vacuum because only there you can detect the emitted Auger electrons. Yes, I recall that.

...

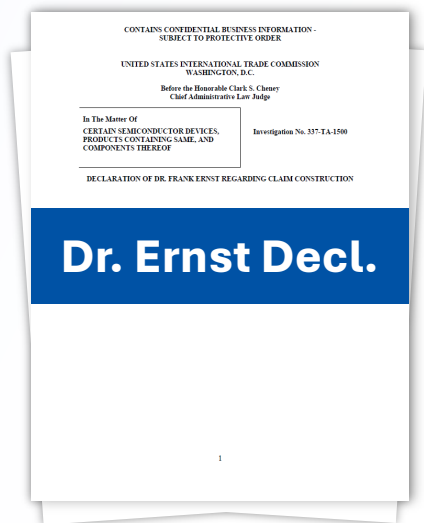
Q. And do you recall providing some testimony about atom probe tomography?

A. I do.

Q. And you referred to that as APT for short?

A. Yes. There is an instrument and a method, and the method is called APT, atom probe tomography. Whereas, the instrument is called a topographical atom probe, TAP.

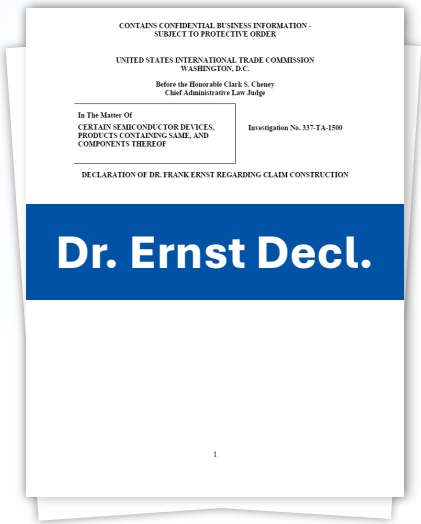
Undisputed Evidence – Yielding Different Results



37. In my opinion, a POSA would not have been reasonably certain as to where the “nickel silicide surface at a silicon interface” is located based on the language of the specification. The prosecution history also fails to provide any guidance. In particular, the '235 patent does not provide any guidance as to how the location of the “nickel silicide surface at a silicon interface” is to be *measured*.

38. A number of acceptable methods of “microcharacterization” are in use for this purpose, but they yield *different* results. For example, researchers have used XEDS (X-ray Energy-Dispersive Spectrometry) in conjunction with SEM (Scanning Electron Microscopy) to measure profiles $X_i[z]$ or $W_i[z]$ of the atom fraction X or the mass fraction W of elements $i, i = 1 \dots N$, and determine surface/interface positions from the characteristics of the profiles.

Undisputed Evidence – Yielding Different Results



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• • •

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58. A person of skill in the art would understand that applying these different methodologies can – *will* – lead to different results.

• • •

58. A person of skill in the art would understand that applying these different methodologies can – *will* – lead to different results.

Undisputed Evidence – Yielding Different Results

Dr. Ernst

Tower Semiconductor
Technical Expert

- Q. Right. So in your declaration, you provide various techniques for measuring material composition, and those techniques will yield different results for the silicon composition at a particular location, correct?
- A. That is correct, but what is of greater interest to us here is that the uncertainty in composition will provide uncertainty in location.

Undisputed Evidence – Yielding Different Results

Dr. Ernst

Tower Semiconductor
Technical Expert

- Q. And would these methodologies yield different results for the location of the silicon interface if tested in a real product?
- A. Yes. If you provide a definition of silicon interface that is tied to a certain silicon level. Like you say, the silicon interface is where the silicon level has reached 90 percent coming from the silicide side. That—the location where 90 percent silicon are reached would come out differently if you compare the line scans from two different methods.

And as I said earlier, strictly speaking, they will evenly come out differently if you do the same linescan with the same instrument twice under the same parameters because there are statistical fluctuations.

Undisputed Evidence – Yielding Different Results

Dr. Ernst

Tower Semiconductor
Technical Expert

Q. And in your view, if the term “silicon interface” required a particular silicon composition, then the various techniques described in your declaration might yield—will yield a different silicon concentration rendering the term indefinite, fair?

A. It will yield different locations. So different techniques will yield different profiles, and in these different profiles, the location where silicon is 80 percent will be different. One—

Q. Okay. Right. So if—

A. If I can add one more thing.

Q. Yep.

A. If you have fluctuations, statistical fluctuations, if you just look at the figure that we looked at with the lines A and B, these lines that you see, they go up and down. So it could be that the 80 percent silicon level is hit twice at two different locations (witness indicating).

Which one would you then choose to be the silicon interface?

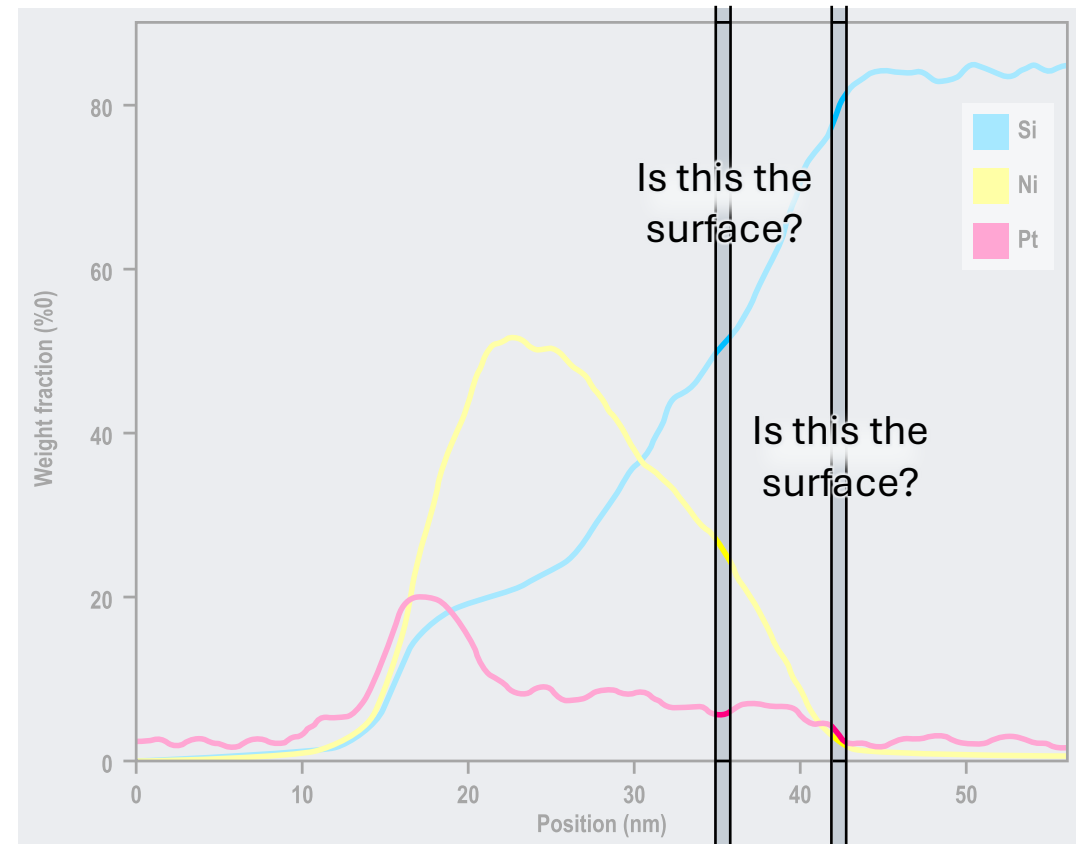
Evidence Confirms a POSA Could Arbitrarily Select a Silicon Concentration

Claim 1[d]: wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface.

sharp interface will be *diffuse*. That means it will no longer be straight forward to assign an exact location to the interface, unless by an arbitrary criterion. For the concrete case of Ni–Pt in contact with Si, Ni and Pt will both diffuse into Si. The exact location of “a silicon interface,” as conceived in the '235 patent, is no longer meaningful unless determined by an arbitrary criterion. The '235 patent, however, does not provide any such criterion.

...

53. For example, a paper entitled “Silicide-phase evolution and platinum redistribution during silicidation of $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ specimens” by Praneet Adusumilli, David N. Seidman, and Conal E. Murray (“Adusumilli”) (Ex. 17) explores the Si/silicide interface using Atomic Probe Tomography (APT). Ex. 17 at 1. Using APT, the authors obtained highly accurate topographical 3D maps of *arbitrarily* chosen Si reference levels, e.g. $X_{\text{Si}} = 0.75$, a “75 at. % Si



Complainant's Expert Has No Opinion

Dr. Liu

GlobalFoundries
Technical Expert

Q. What different microcharacterization methodologies are you familiar with?

A. I believe I'm familiar with a lot of them. But, again, I have to say that in forming my opinion in this declaration for the claim—purpose of claim construction, I did not need to rely on any techniques or—or—or detailed knowledge of any techniques trying to form my opinion for this—for this declaration.

Complainant's Expert Has No Opinion

Dr. Liu

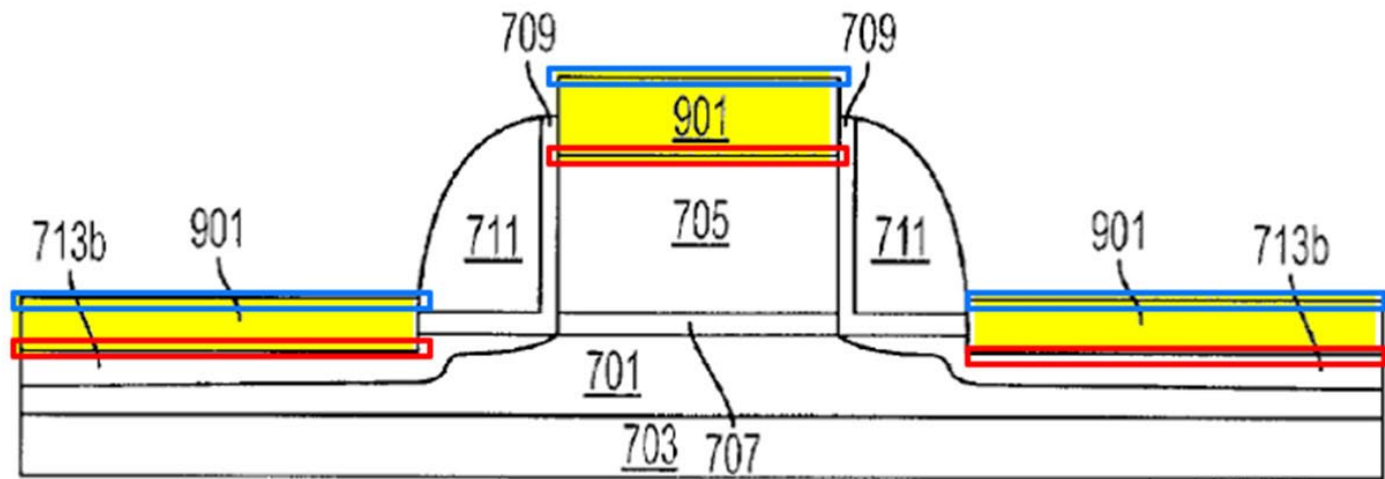
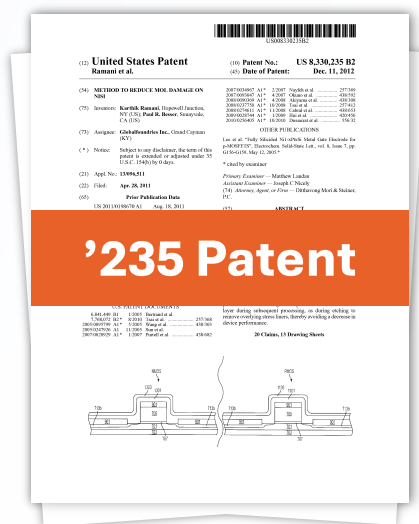
GlobalFoundries
Technical Expert

Q. Do you agree or disagree that one of the microcharacterization methodologies that a person of skill in the art can use in order to perform an analysis to determine the location of that silicon -- silicon interface -- is atom probe tomography? Do you agree or disagree?

A. Again, sitting right here, I did not offer an opinion on how or what technique to use to determine, and I have not done that analysis, so I cannot comment on that.

The Specification and Prosecution History Provide No Guidance

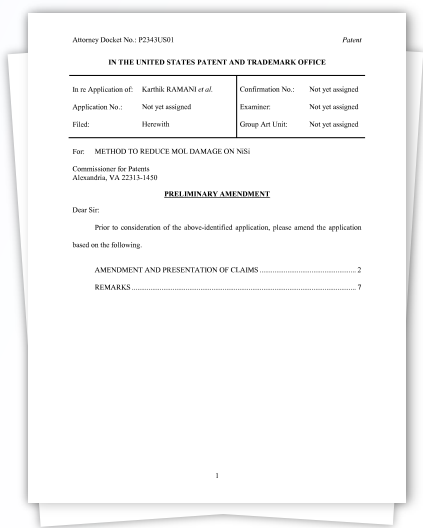
as illustrated in FIG. 9. The resulting NiSi layers **901** have a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the bottom surface that contacts the silicon gate electrodes **705** and source/drain regions **713b**, to about 30 wt. % to about 70 wt. %, such as about 50 wt. %, at the opposite surface. NiSi



The Specification and Prosecution History Provide No Guidance

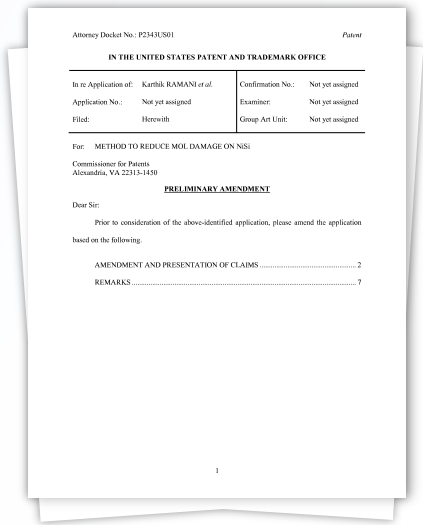
Telephonic Interview

Applicants express appreciation for Examiner Nicely’s courtesy in granting and conducting a telephonic interview on July 24, 2012. During the interview, a proposed amendment which has been implemented herewith and the rejections under 35 U.S.C. § 102(b) and 103 were discussed. In addition during the interview, the Examiner courteously suggested reciting the platinum content at the silicon interface, which has also been incorporated in this amendment. The Examiner agreed during the interview that the proposed amendment, clarifying that the platinum composition gradient starts at a nonzero value at the silicon interface and spans the entire depth of the nickel silicide, appears to overcome the rejections relying upon *Okuno* for a platinum gradient. The Examiner, however, indicated that he still needed to update his search.



**'235 Patent
Prosecution History**

The Specification and Prosecution History Provide No Guidance



**'235 Patent
Prosecution History**

14. (Currently Amended) A semiconductor device comprising:

a silicon substrate,

a transistor, comprising source/drain regions and a gate electrode, on the substrate; and

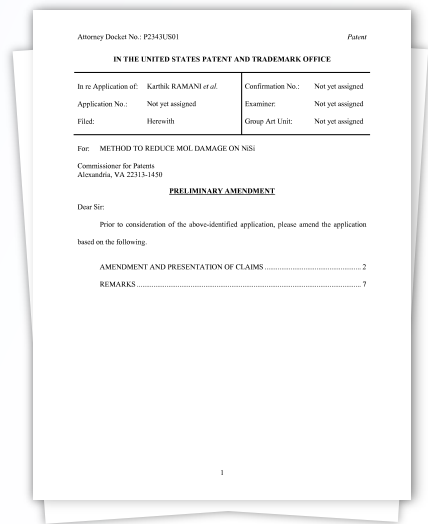
a nickel silicide layer on the gate electrode and source/drain regions,

wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide,

with platinum increasing in amount in a direction away from the source/drain regions and

the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface.

The Specification and Prosecution History Provide No Guidance



'235 Patent
Prosecution History

Substance of Interview

(For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Applicant's representative contacted the Examiner to discuss the Okuno prior art document as well as proposed amendments to the claims. Examiner indicated that the proposed amendments to overcome the art needed to include not only that there is a platinum concentration gradient from the silicon interface with the silicide to the opposite surface of the silicide, the platinum concentration gradient at the silicon interface should be amended into the independent claims since the term gradient can include a zero to non-zero transition because gradient merely means a change in value of a quantity (platinum concentration) over a variable (depth within gate electrode/nickel silicide). Since the variable has no defined spacing or increment and change has no defined increment, without introducing that the silicon interface has a non-zero platinum concentration of 8 wt % to 12 wt %, the art could still read on the proposed amendments. However, the inclusion of the 8 wt % to 12 wt % at the silicon interface into the independent claims would overcome Okuno.

“adjacent to”

'983 Patent, Claims 1, 5, 13, 17, and 20

“the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate”

'983 Patent, Claims 1, 13, and 20

Disputed Terms

Disputed Term

“adjacent to”

'983 patent, claims 1, 5, 13,17, & 20

“the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate”

'983 patent, claims 1, 13, & 20

Tower Semiconductor

Proposed Construction

Indefinite

Indefinite

GlobalFoundries U.S.

Proposed Construction

Plain and ordinary meaning; not indefinite.

Plain and ordinary meaning; not indefinite.



(12) **United States Patent**
Zhang et al.

(10) **Patent No.:** **US 8,507,983 B2**
(45) **Date of Patent:** **Aug. 13, 2013**

(54) **HIGH VOLTAGE DEVICE**

(75) Inventors: **Guowei Zhang**, Singapore (SG);
Purakh Raj Verma, Singapore (SG)

(73) Assignee: **GLOBALFOUNDRIES Singapore Pte. Ltd.**, Singapore (SG)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **13/550,571**

(22) Filed: **Jul. 16, 2012**

(65) **Prior Publication Data**
US 2012/0280318 A1 Nov. 8, 2012

Related U.S. Application Data

(60) Division of application No. 12/500,620, filed on Jul. 10, 2009, now Pat. No. 8,222,130, which is a continuation-in-part of application No. 12/390,509, filed on Feb. 23, 2009, now Pat. No. 8,053,319.

(51) **Int. Cl.**
H01L 29/76 (2006.01)

(52) **U.S. Cl.**
USPC **257/335; 257/E29.256**

(58) **Field of Classification Search**
USPC 257/335, 346, E29.256
See application file for complete search history.

References Cited

U.S. PATENT DOCUMENTS
5,589,414 A 12/1996 Wian et al.
6,759,675 B2 7/2004 Coutak et al.
6,894,350 B2 5/2005 Shimizu et al.
6,979,864 B2 * 12/2005 Negoro et al. 257/335
7,180,132 B2 2/2007 Cai et al.
7,776,700 B2 8/2010 Yang et al.
7,829,402 B2 * 11/2010 Matocha et al. 438/197
7,935,992 B2 * 5/2011 Chang 257/288
2009/0059111 A1 3/2009 Jang

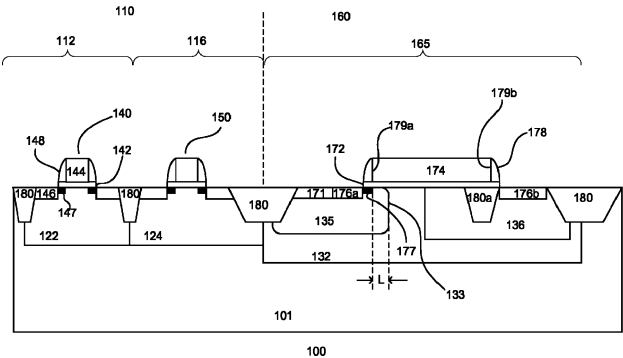
OTHER PUBLICATIONS
Leonard Rubin et al, Narrow n+/p+ isolation in retrograde well implants, Solid State Technology Magazine, Jul. 1, 2003, pp. 119-125, vol. 46 Issue 7, Solid State Technology, USA.

* cited by examiner

Primary Examiner — Quoc Hoang
(74) **Attorney, Agent, or Firm** — Horizon IP Pte Ltd

(57) **ABSTRACT**
A device is disclosed. The device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. A doped drift well adjacent to a second edge of the gate is also included.

20 Claims, 29 Drawing Sheets



Patent No.: **US 8,507,983 B2**

Date of Patent: **Aug. 13, 2013**

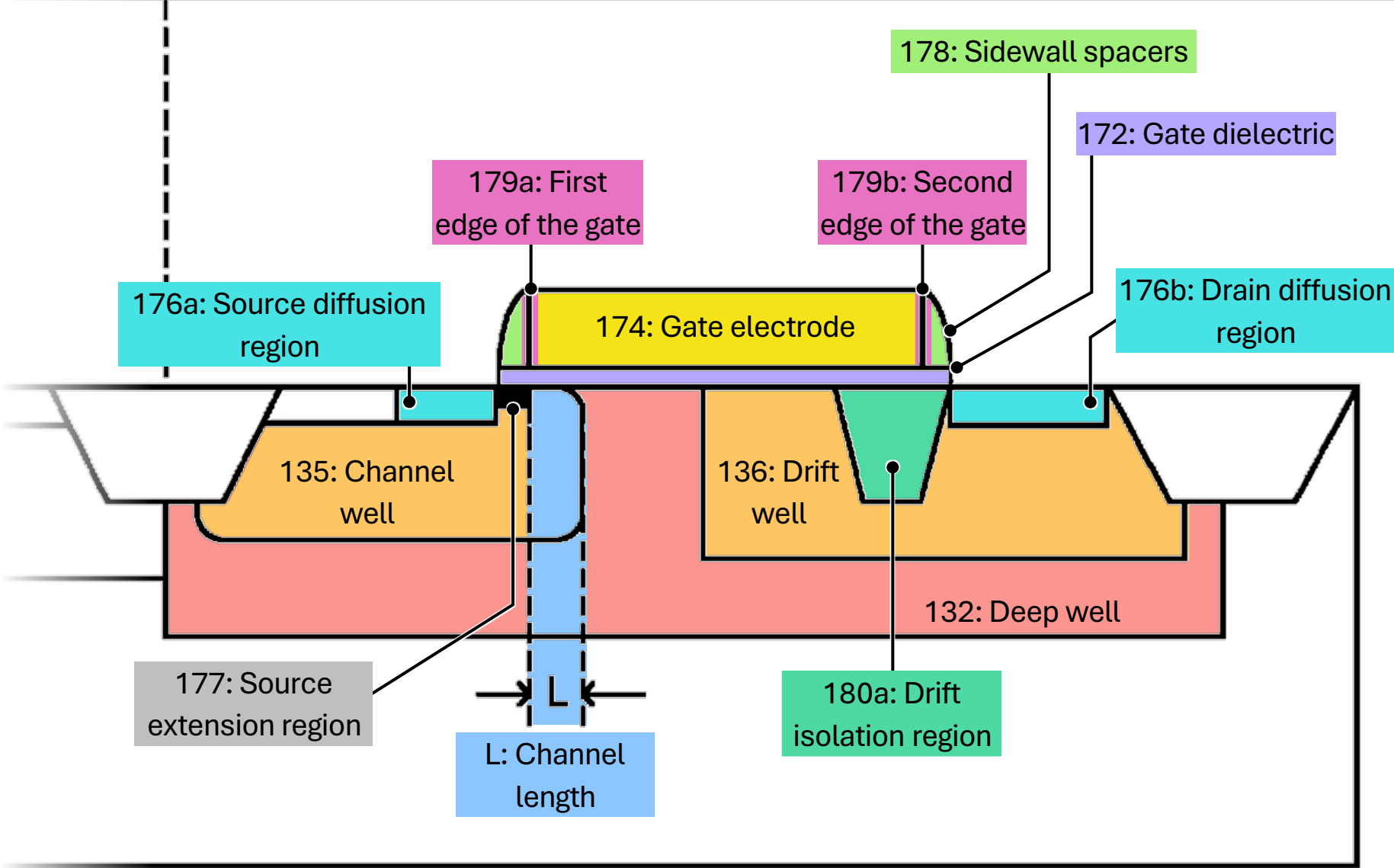
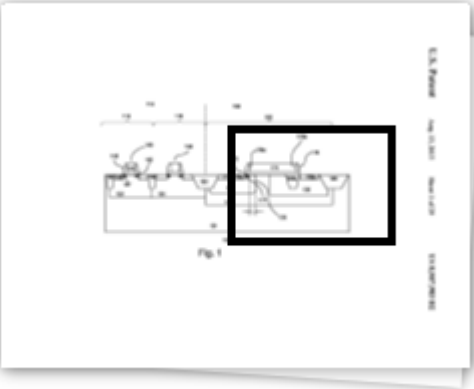
HIGH VOLTAGE DEVICE

Inventors: **Guowei Zhang**, Singapore (SG);
Purakh Raj Verma, Singapore (SG)

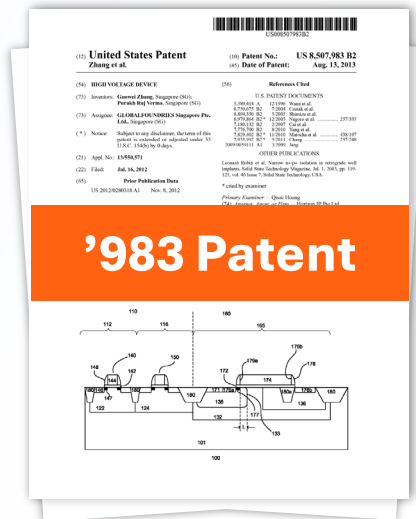
Assignee: **GLOBALFOUNDRIES Singapore Pte. Ltd.**, Singapore (SG)

Filed: **Jul. 16, 2012**

'983 Patent, Figure 1



“adjacent to” Refers to Different Spatial Relationships

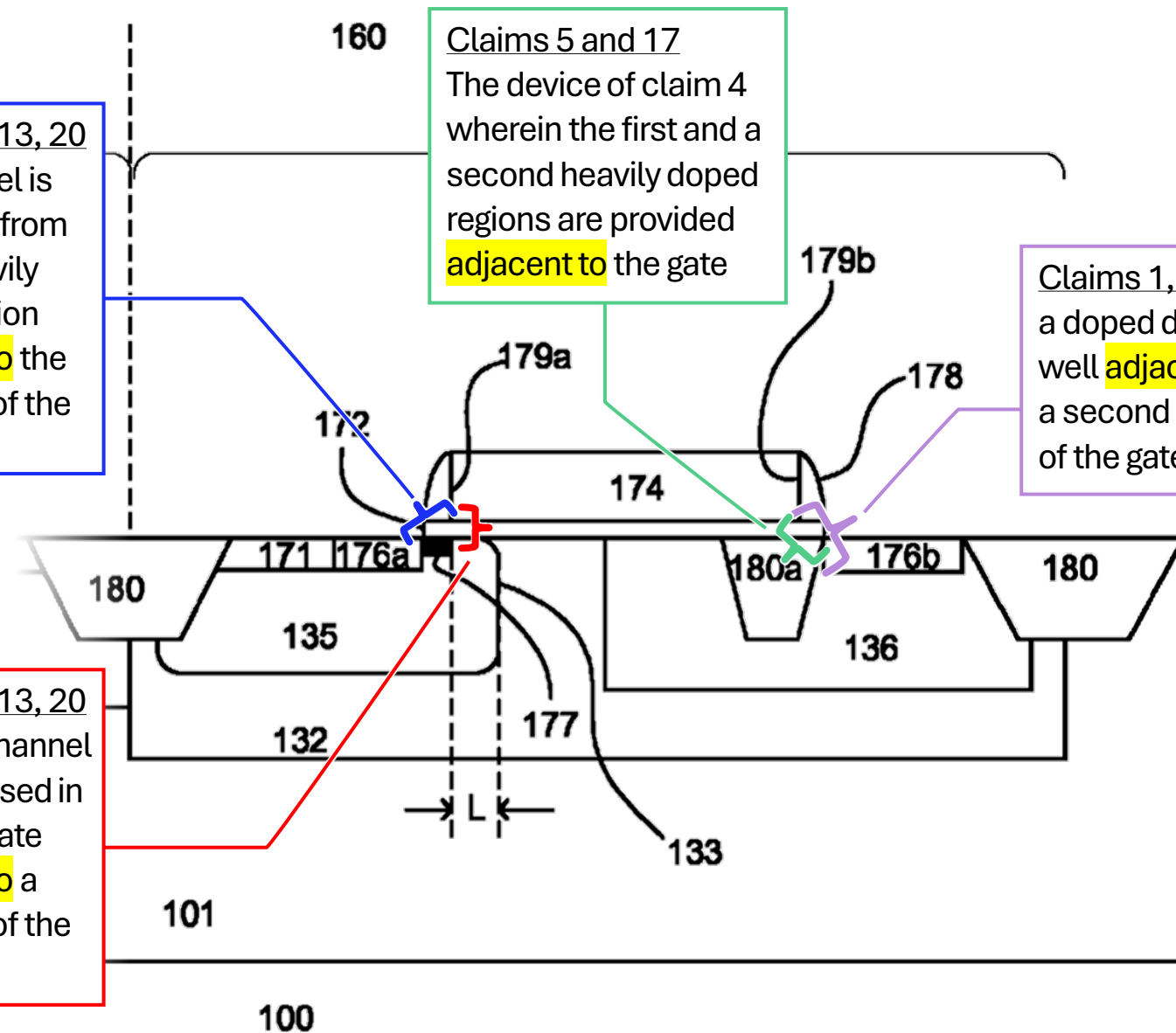


Claims 1, 13, 20
the channel is displaced from a first heavily doped region adjacent to the first edge of the gate

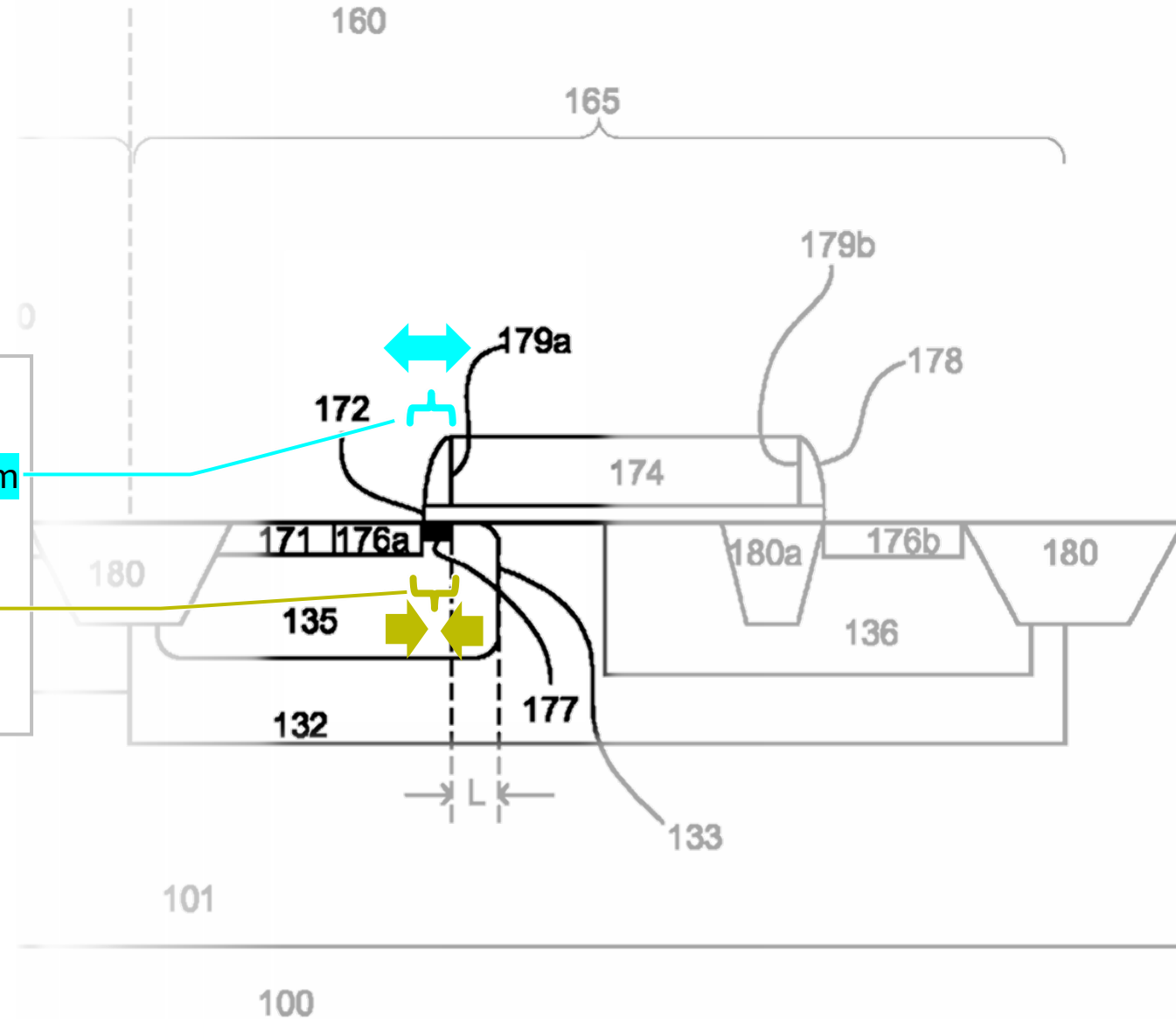
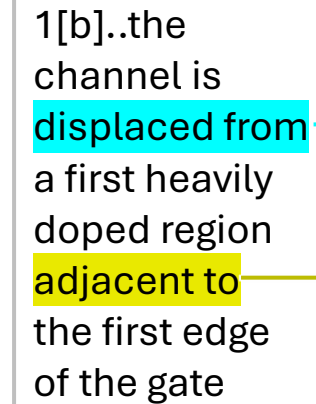
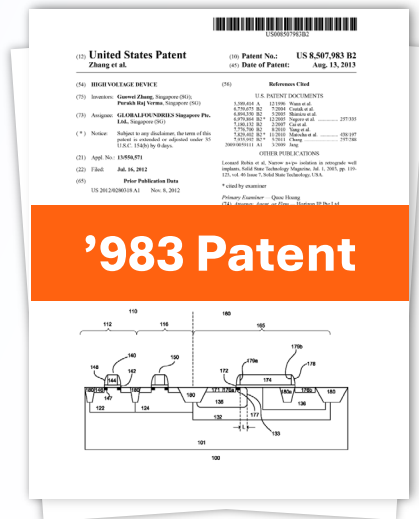
Claims 5 and 17
The device of claim 4 wherein the first and a second heavily doped regions are provided adjacent to the gate

Claims 1, 13, 20
a doped drift well adjacent to a second edge of the gate

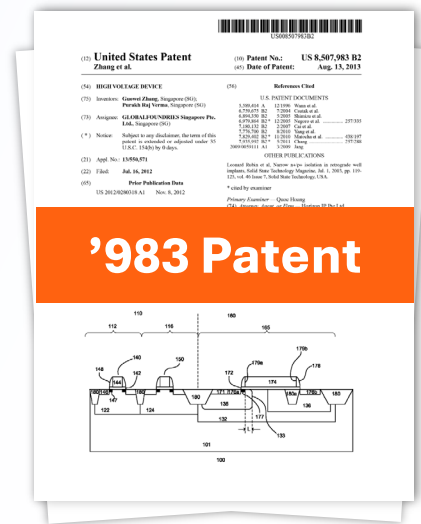
Claims 1, 13, 20
a doped channel well disposed in the substrate adjacent to a first edge of the gate



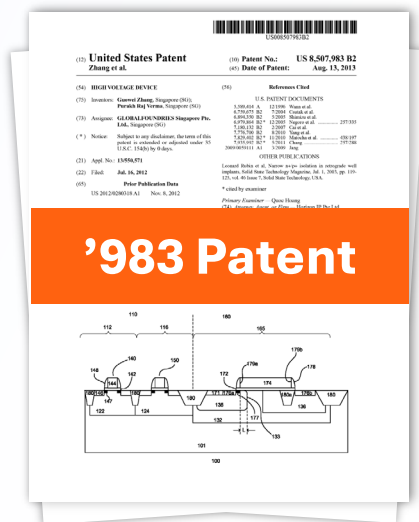
“adjacent to” and “displaced from” Refer to Opposing Spatial Relationships



The Specification Provides No Guidance

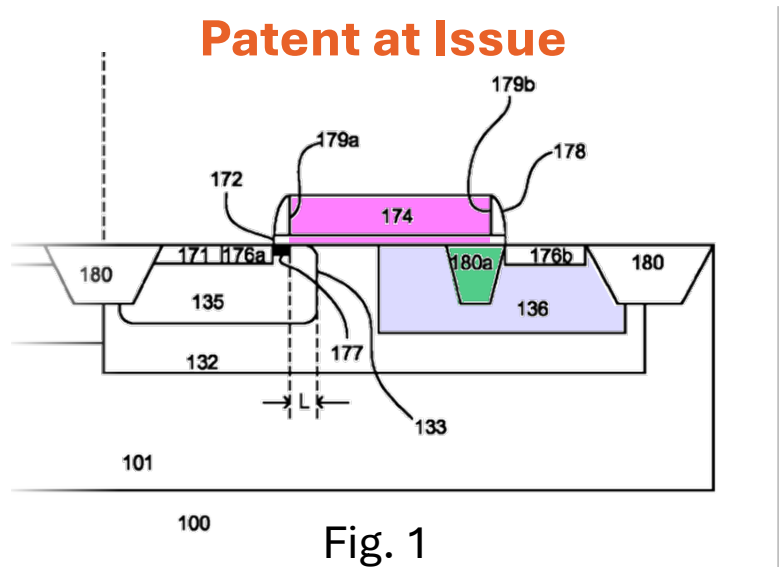


The Specification Confirms That Positioning Can Vary

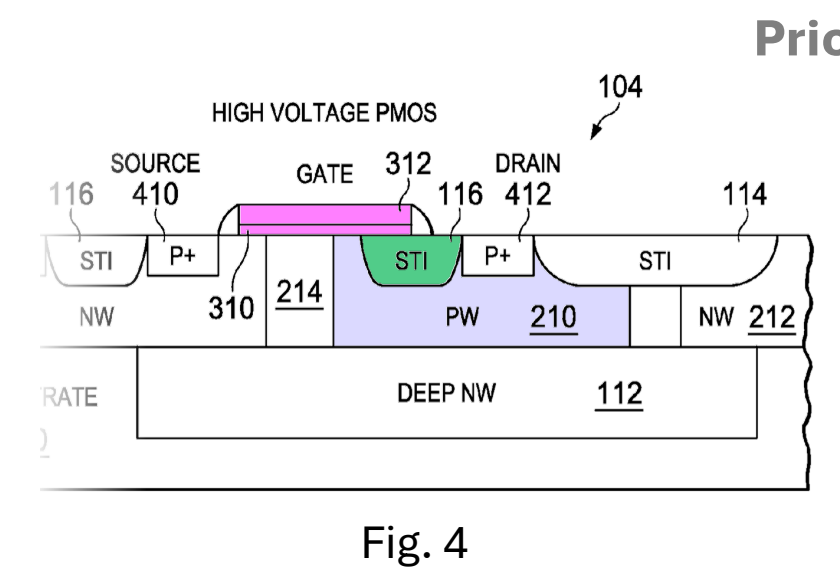


In FIG. 2c, various layers of the gate are formed on the substrate. In one embodiment, a gate dielectric layer 242 is formed over the substrate. The gate dielectric layer, for example, comprises silicon oxide. Other types of dielectric materials, such as silicon oxynitride, may also be useful. Alternatively, high-k, low-k or composite of dielectric materials may be used. The thickness of the gate dielectric layer 242 may be about 30 Å for 1.8V gate voltage transistors. Other thicknesses may also be useful. The thickness, for example, may depend on the gate voltage applications. For example, higher gate voltages may require thicker gate dielectric layers. In one embodiment, the gate dielectric layer is formed by thermal oxidation. Other techniques, such as chemical vapor deposition (CVD), can also be used to form the gate dielectric layer.

GF Contends the Prior Art Does Not Disclose “Adjacent” Elements, Even Though the Prior Art Discloses Distances **Just Like The '983 Patent**

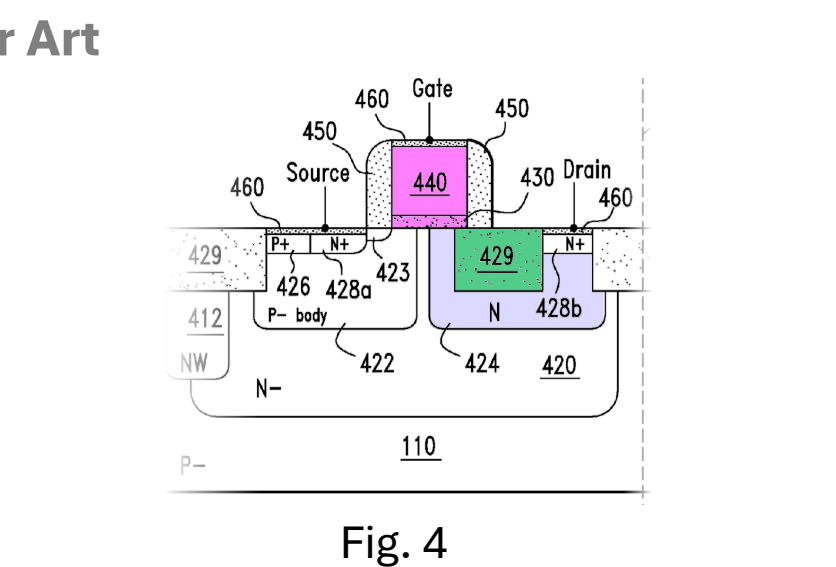


1. A device comprising:
- a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;
 - a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; and
 - a doped drift well adjacent to a second edge of the gate.



GF Validity Contentions

element 210 is a “doped channel well” as claimed. For example, Tower has not identified a cited passage in Wu ’291 establishing that element 210 (the mapped “doped drift well”) is positioned “adjacent to a second edge of the gate” as the asserted limitation requires. Wu ’291’s disclosed structure shows element 116 (field oxide) between the gate edge and the drain region, and Tower has not established the asserted adjacency relationship between the drift well and the gate’s second edge from Wu ’291’s cited passages.



GF Validity Contentions

For example, Tower has failed to identify the claimed displacement, or any spacer that defines the alleged displacement. For example, Tower has failed to explain how element 424 comprises the claimed doped drift well adjacent to a second edge of the gate. Tower maps the asserted “drift isolation region” to element 429 and the asserted “doped drift well” to element 424,

GF's Expert: There Is No "definite requirement" for the Scope of "adjacent"

Dr. Liu
GlobalFoundries
Technical Expert

11 Q No problem.

12 What distance can separate the second edge
13 of the gate and the drift well and those two
14 things still be adjacent to each other?

15 MR. AL-MARZOOG: Object to the form.
16 Incomplete hypothetical.

17 THE WITNESS: I think the -- there's no
18 definite -- definite requirement. You have to
19 take this "adjacent" term in its proper context.

GF's Shifting Sands Approach Supports Indefiniteness

Interval Licensing LLC v. AOL, Inc.



“As we have explained, a term of degree fails to provide sufficient notice of its scope if it depends on the unpredictable vagaries of any one person’s opinion.”

Interval Licensing LLC v. AOL, Inc., 766 F.3d 1364, 1371 (Fed. Cir. 2014)

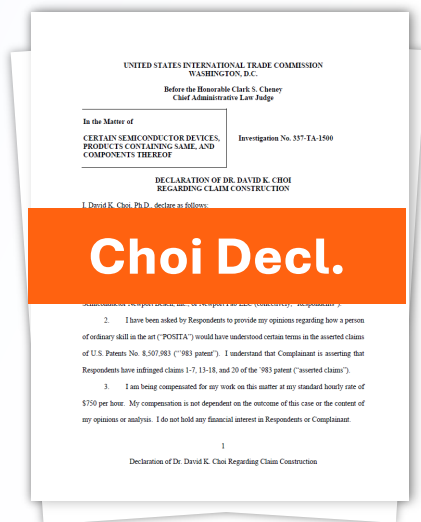
Icon Health & Fitness, Inc. v. Polar Electro Oy

Terms are indefinite where “the scope could vary from day-to-day and from person-to-person.”



Icon Health & Fitness, Inc. v. Polar Electro Oy, 656 F. App'x. 1008, 1016 (Fed. Cir. 2016)

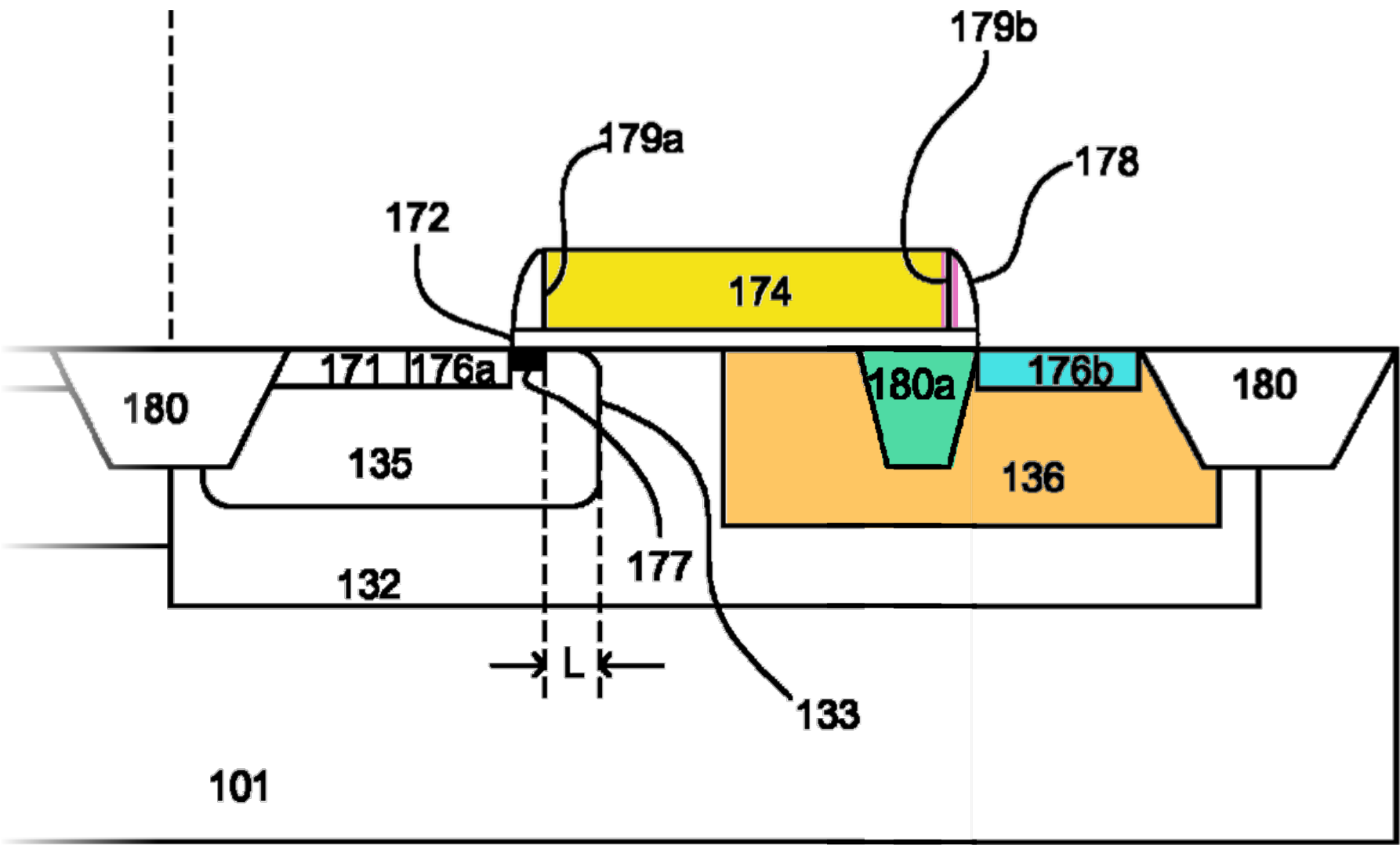
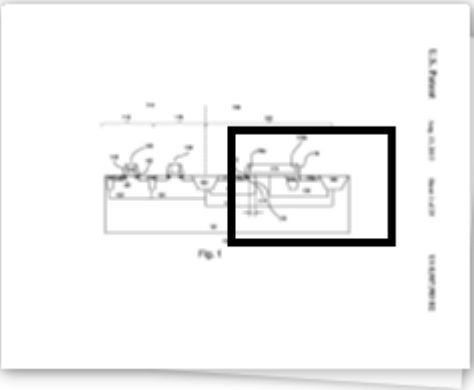
Dr. Choi: POSITA Would Know Spacing Between “Adjacent” Elements Can Vary



Accordingly, the spatial relationship between the doped drift well and the second edge of the gate may vary based on the spatial dimensions of the drift isolation region. A POSITA would have understood, and the specification of the '983 patent confirms, that such spatial dimensions and relationships are highly dependent on the design of a given device and its required specifications and application. (*E.g.*, '983 patent, 5:15-24 (“Disposing the drift well at other positions, such as partially underlying the gate may also useful.”), 7:8-13.)

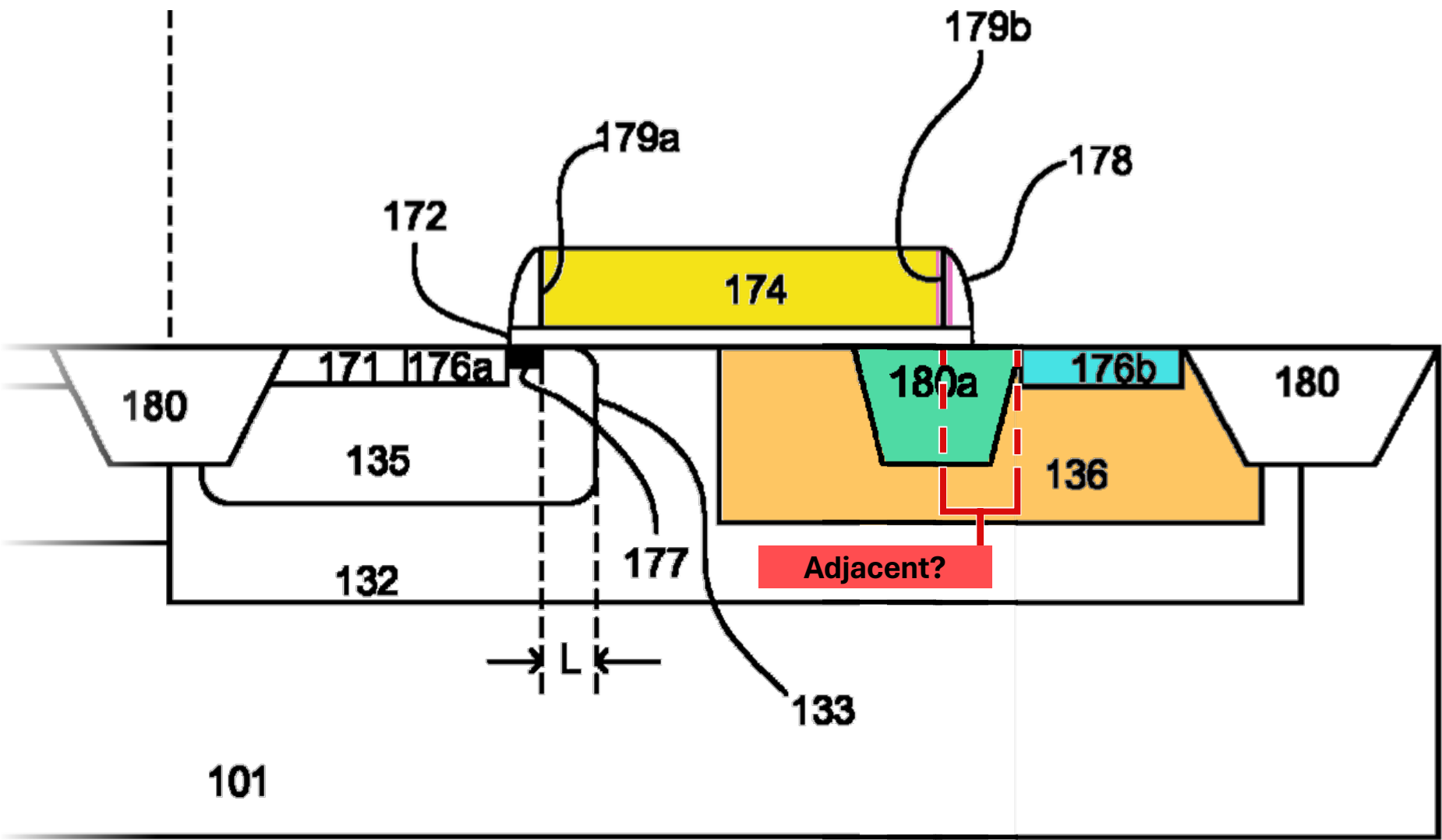
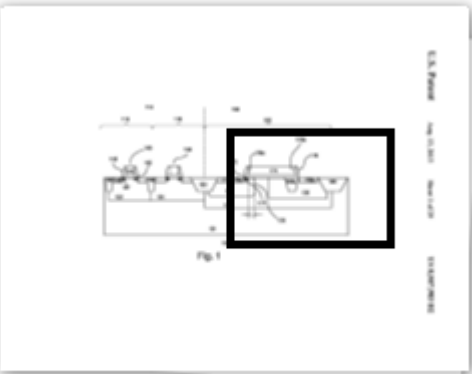
Spatial Relationships May Vary Based on, e.g., the Size of the Drift Isolation Region

Claim 1[c]: ...“a doped drift well adjacent to a second edge of the gate.”



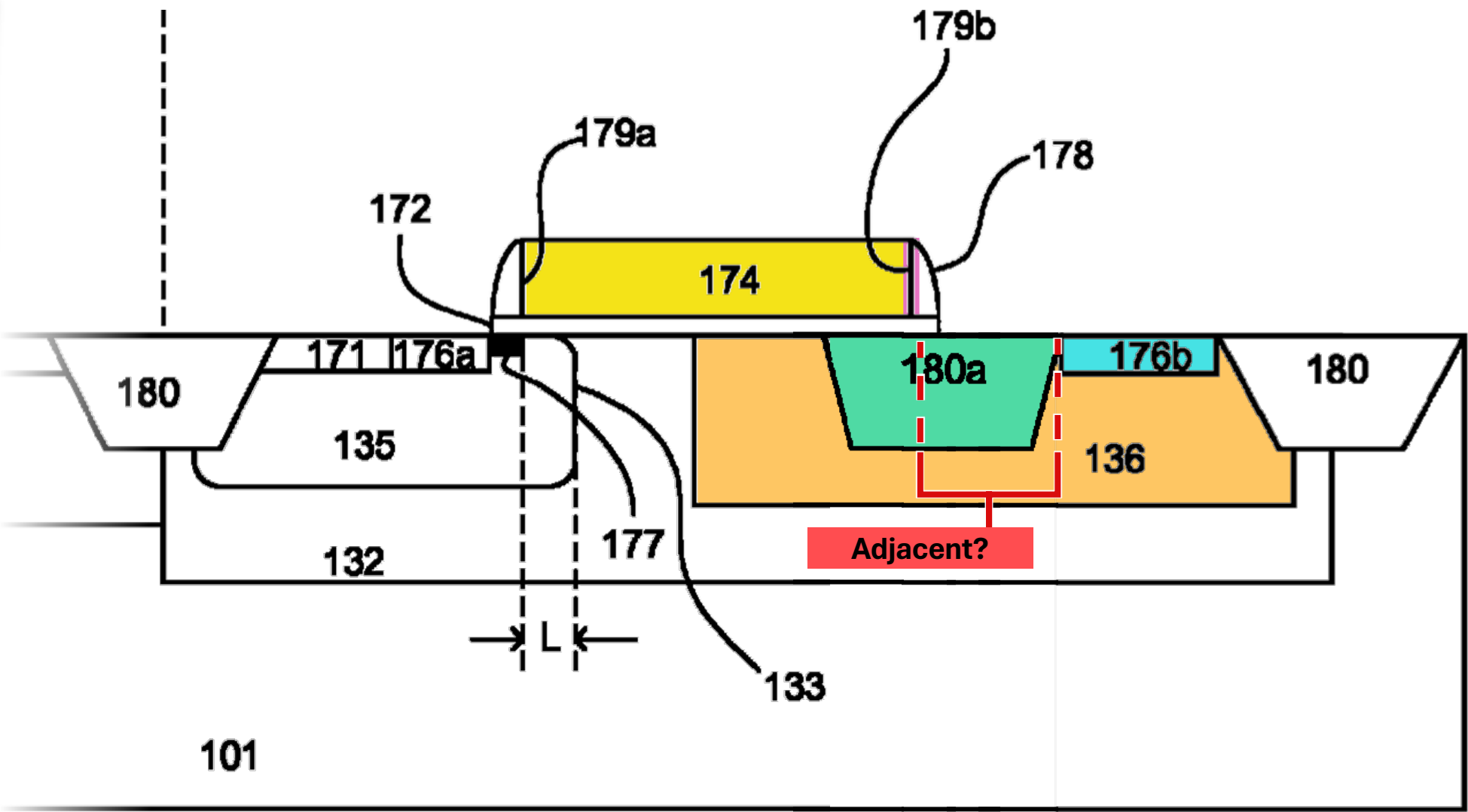
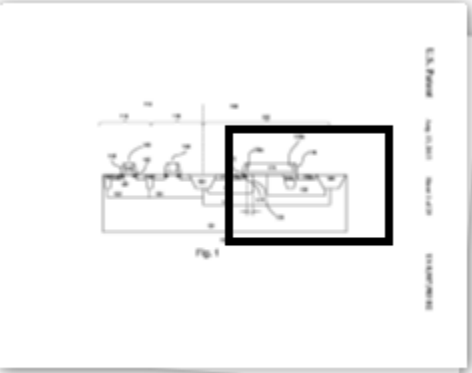
Spatial Relationships May Vary Based on, e.g., the Size of the Drift Isolation Region

Claim 1[c]: ...“a doped drift well adjacent to a second edge of the gate.”

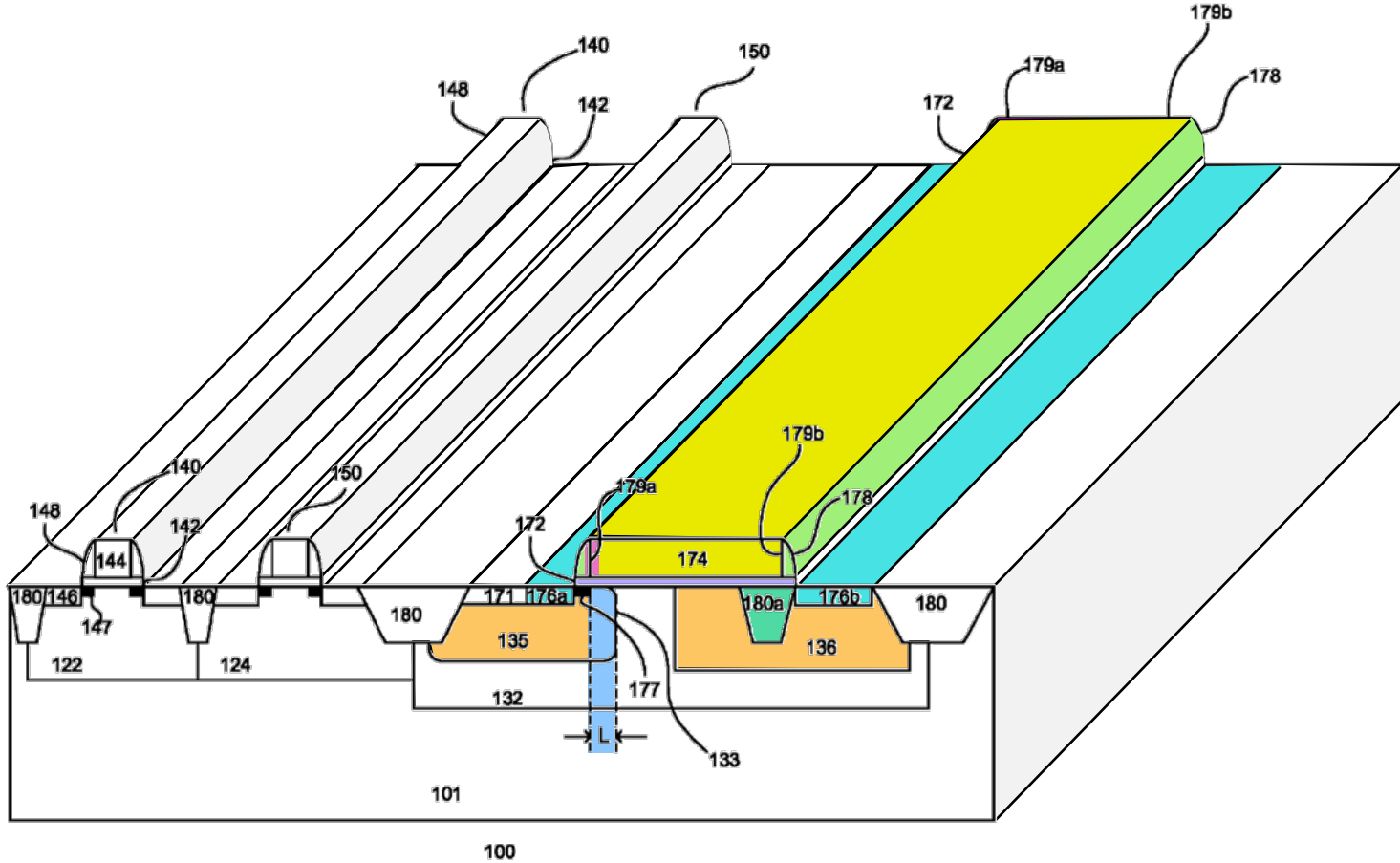
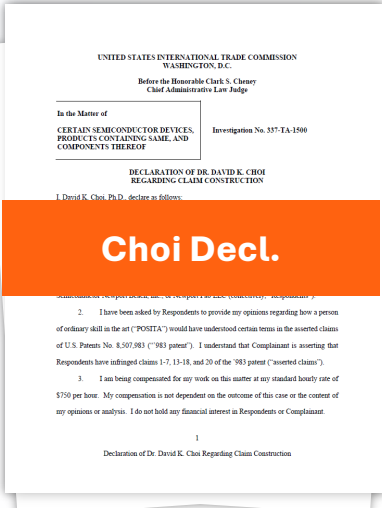


Spatial Relationships May Vary Based on, e.g., the Size of the Drift Isolation Region

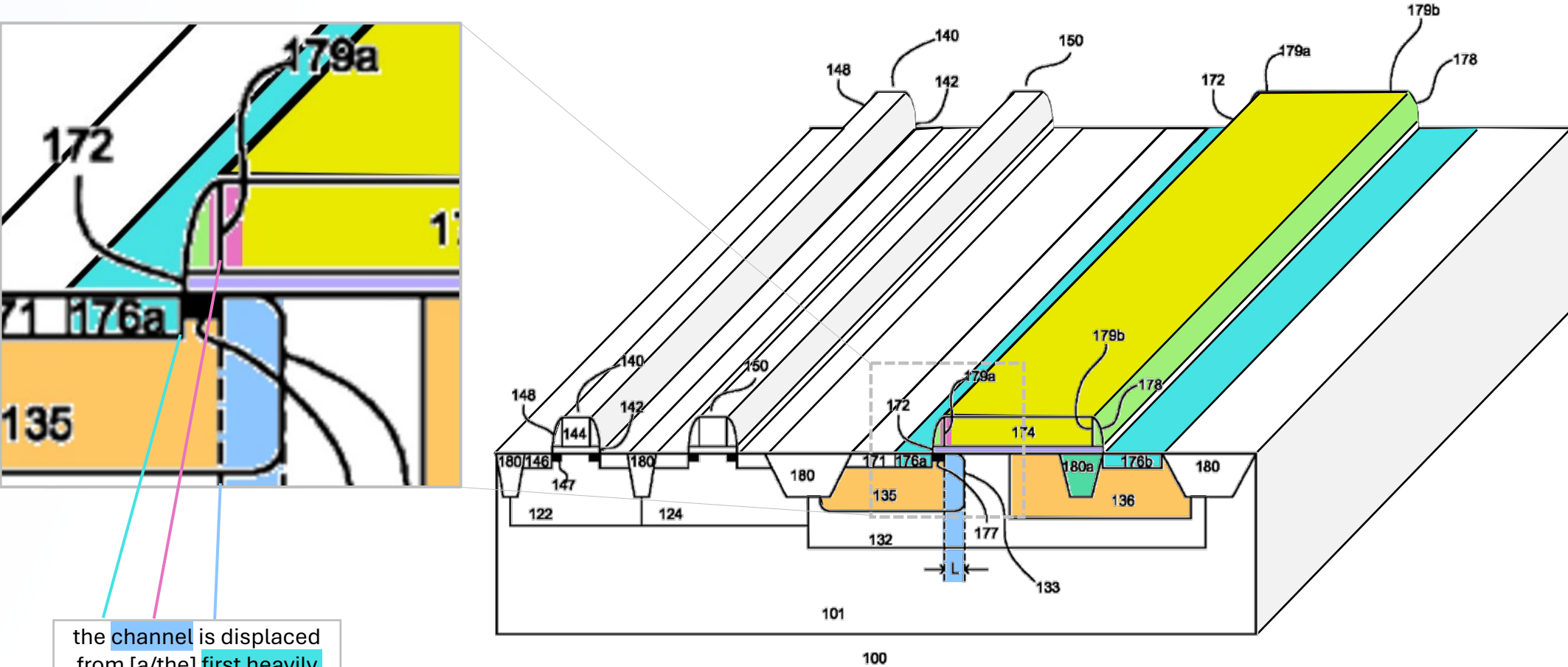
Claim 1[c]: ...“a doped drift well adjacent to a second edge of the gate.”



Dr. Choi: A POSITA Would Understand the Claimed Device is in Three Dimensions

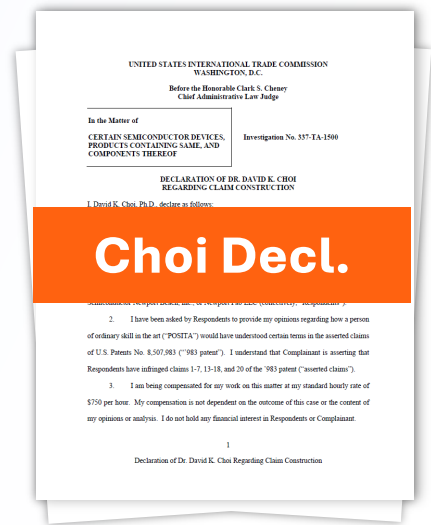


Dr. Choi: A POSITA Would Understand the Claimed Device is in Three Dimensions



the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate

Dr. Choi: POSITA Would Have No Understanding of Order of Magnitude of “adjacent”



In the three-dimensional world in which these devices actually exist, a POSITA would not have had any reasonable certainty in terms of what point to measure the claimed “adjacen[cy]” relationship when looking at the respective three-dimensional structures. Given this reality, a POSITA would not have understood whether the term “adjacent to” would be satisfied by a picometer, a nanometer, a micrometer (microns), 10’s of microns, 100’s microns, etc.

**“the channel having a channel length that
is controlled by ion implantation”**

'983 Patent, Claims 7 and 18

Disputed Term

Disputed Term

“the channel having a channel length that is controlled by ion implantation”

'983 Patent, Claims 7 & 18

Tower Semiconductor

Proposed Construction

“the channel having a channel length that is formed by implanting ions into the substrate to form a channel well”

GlobalFoundries U.S.

Proposed Construction

Plain and ordinary meaning

Formation of Channel Well 135

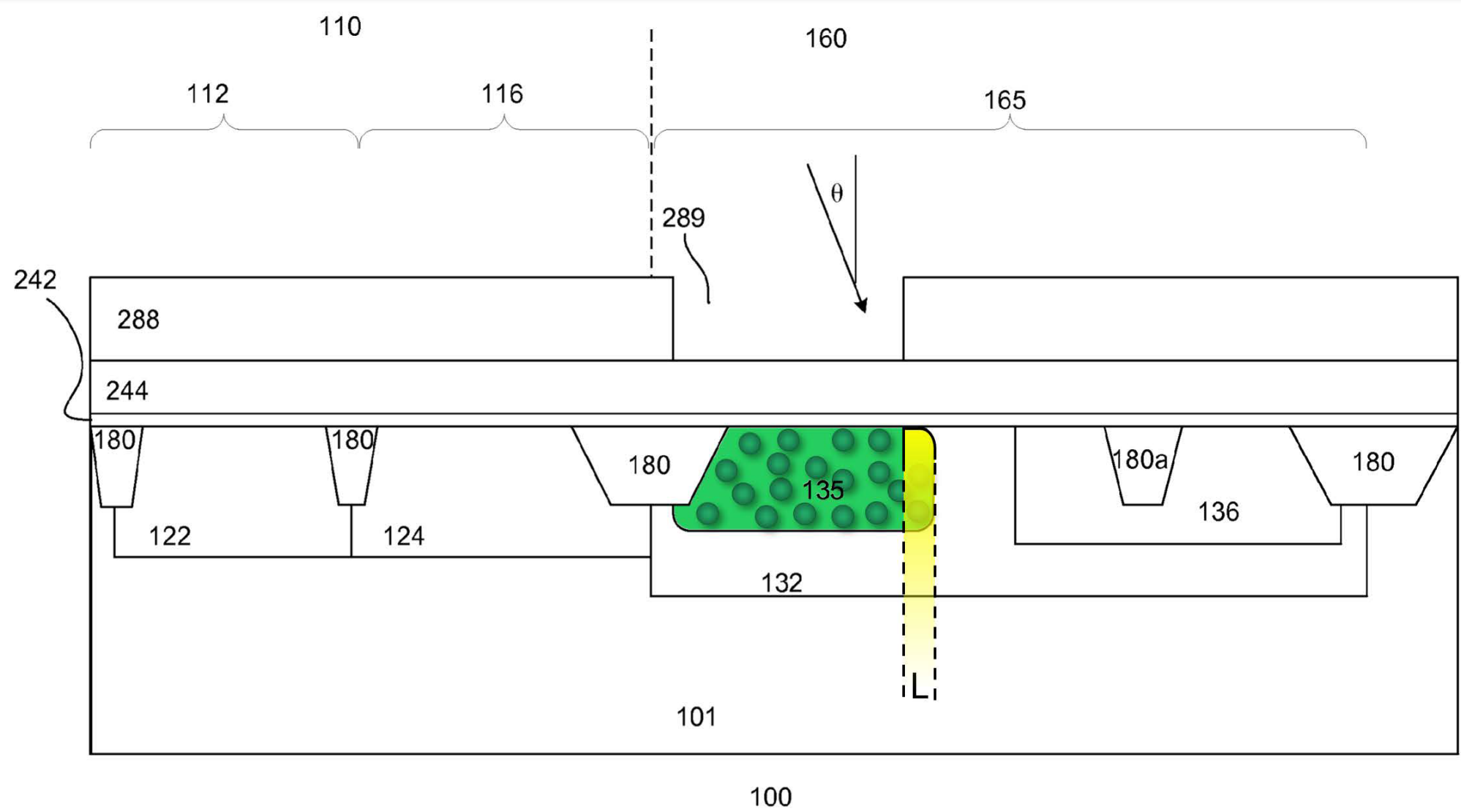
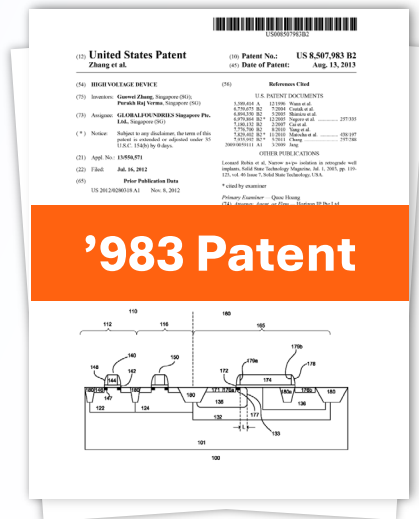


Fig. 2e

Ion Implantation Controls Channel Length Through Formation of the Channel Well



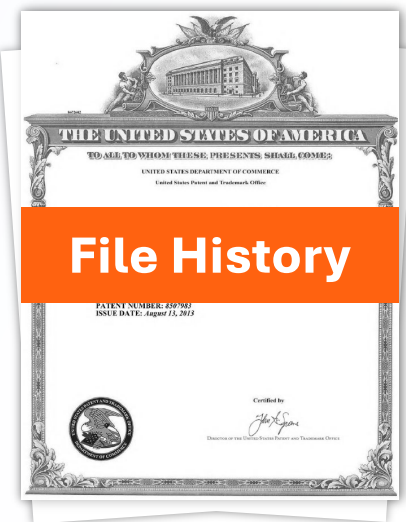
File History: Channel Length Controlled by Ion Implantation Is Met by Channel Well

Original claim

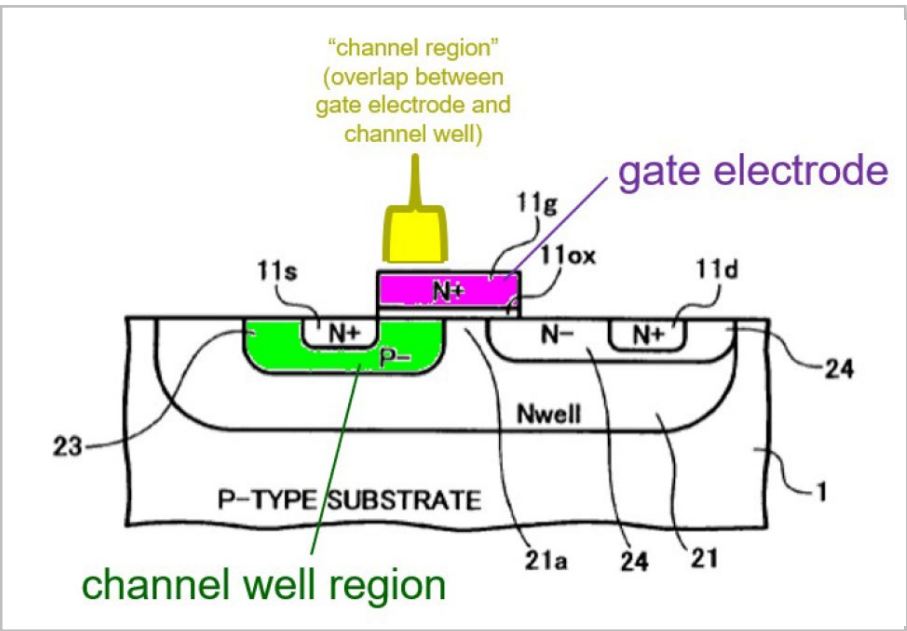
7. The device of claim 1 wherein the effective channel length is controlled by ion implantation.

Examiner finding in 10/26/12 non-final rejection

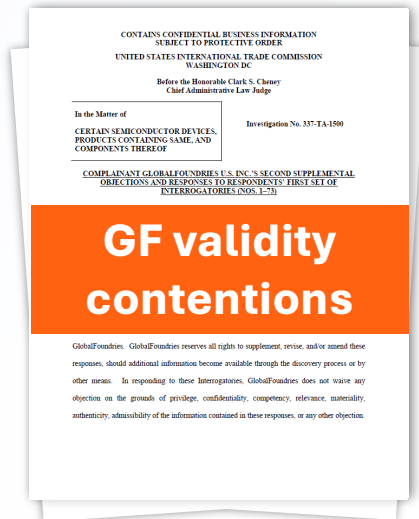
Regarding claims 7-12, Negro teaches the channel length (col. 15, lines 19-21, and Fig. 1).



Negoro prior art

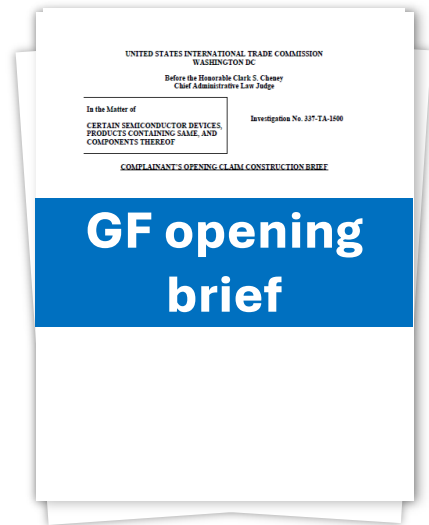


In an Effort to Distinguish Prior Art, GF Argues Forming a Channel Well Is Not Enough

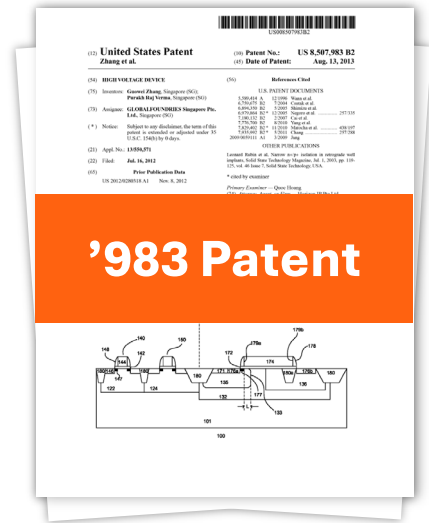


Tower has separately failed to meet its burden to establish by clear and convincing evidence that any charted reference discloses the asserted limitation that the channel have a channel length that is controlled by ion implantation. Tower's charts recite this limitation but do not identify, for each charted reference, a cited passage establishing that the length of the channel—as distinct from the formation of some doped region by implantation—is controlled by an implantation parameter. For example, Lee '935 describes the channel as formed “between the first

GF's Purported Distinction Between “controlling” and “forming” Is Wrong



The claim language and specification refute Respondents’ proposed construction. The claim language uses the term “controlled by,” not “formed by.” The specification uses both terms separately: while components (such as the “channel”) are “formed,” dimensions of such components (such as the “channel length”) are “controlled.” ’983 patent, 8:56-57 (“the substrate is implanted with dopants *to form* a channel well 135.”), 5:57-59 (“The effective channel length, in one embodiment, is *controlled by* ion implantation”). It makes little sense to “form” a “channel length” as Respondents propose.



channel length. Process variations in process overlay requires large process windows. This makes forming short channel lengths to achieve low RDs_{on} difficult if not impossible. Fur-

“edge of the gate”

'983 Patent, Claims 1,2,6,13–17, and 20

Disputed Term

Disputed Term

“edge of the gate”

'983 Patent, Claims 1, 2, 6, 13–15, & 20

Tower Semiconductor

Proposed Construction

“edge of the gate electrode
over the gate dielectric”

GlobalFoundries U.S.

Proposed Construction

Plain and ordinary meaning

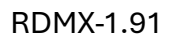
Tower

2 3

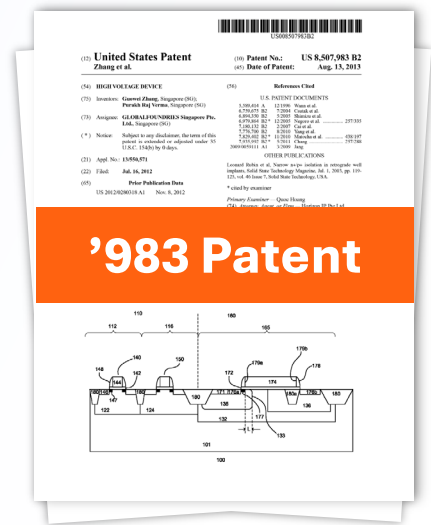
GF

?





Claims: A Gate Includes a Gate Electrode Over a Gate Dielectric



1. A device comprising:
a substrate prepared with an active device region, wherein
the active device region includes a gate, the gate includ-
ing a gate electrode over a gate dielectric;

...

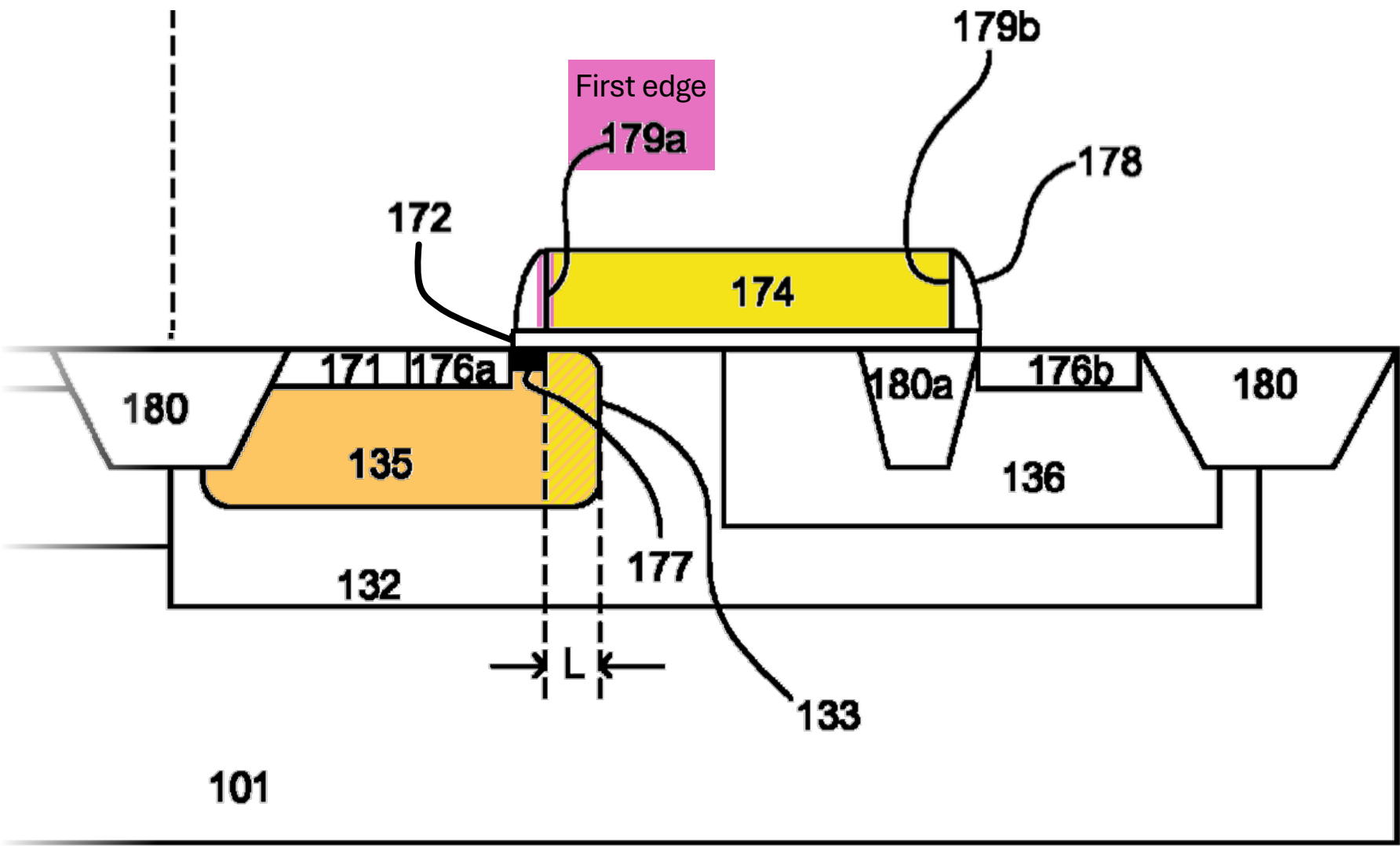
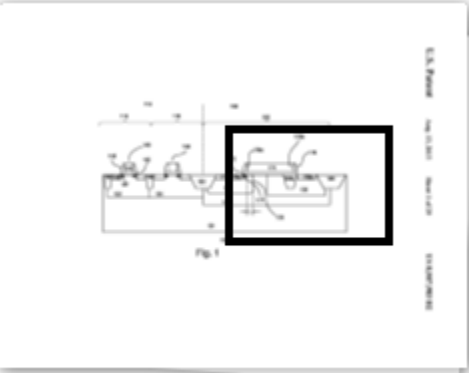
13. A device comprising:
a substrate prepared with an active device region, wherein
the active device region includes a gate, the gate includ-
ing a gate electrode over a gate dielectric;

...

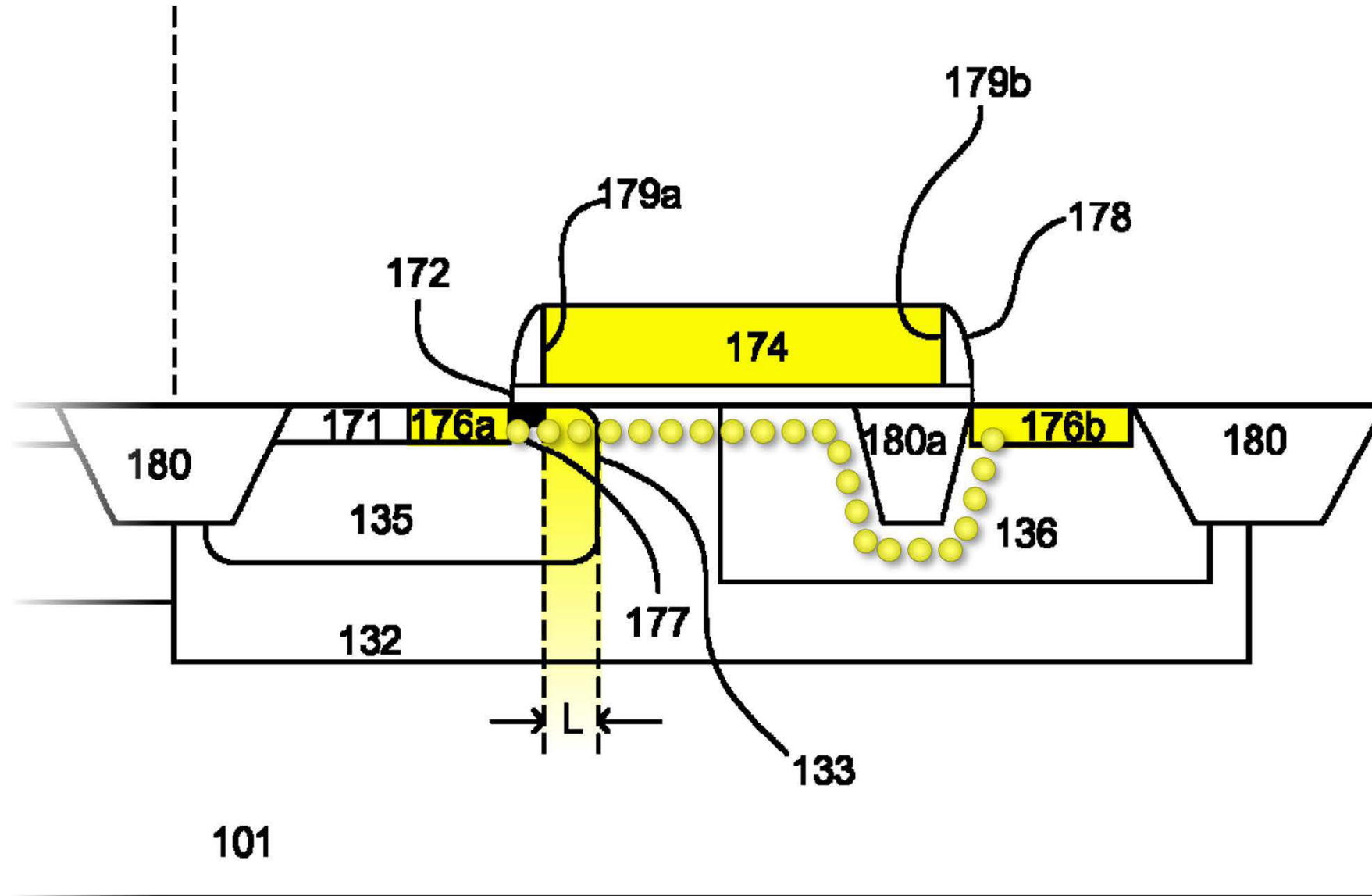
20. A semiconductor device comprising:
a substrate prepared with an active device region, wherein
the active device region includes a gate, the gate includ-
ing a gate electrode over a gate dielectric;

Claims: Edge of the Gate Electrode Over the Gate Dielectric Defines the Channel

Claim 1[b]: “the first edge of the gate and channel edge beneath the gate define a channel of the device”

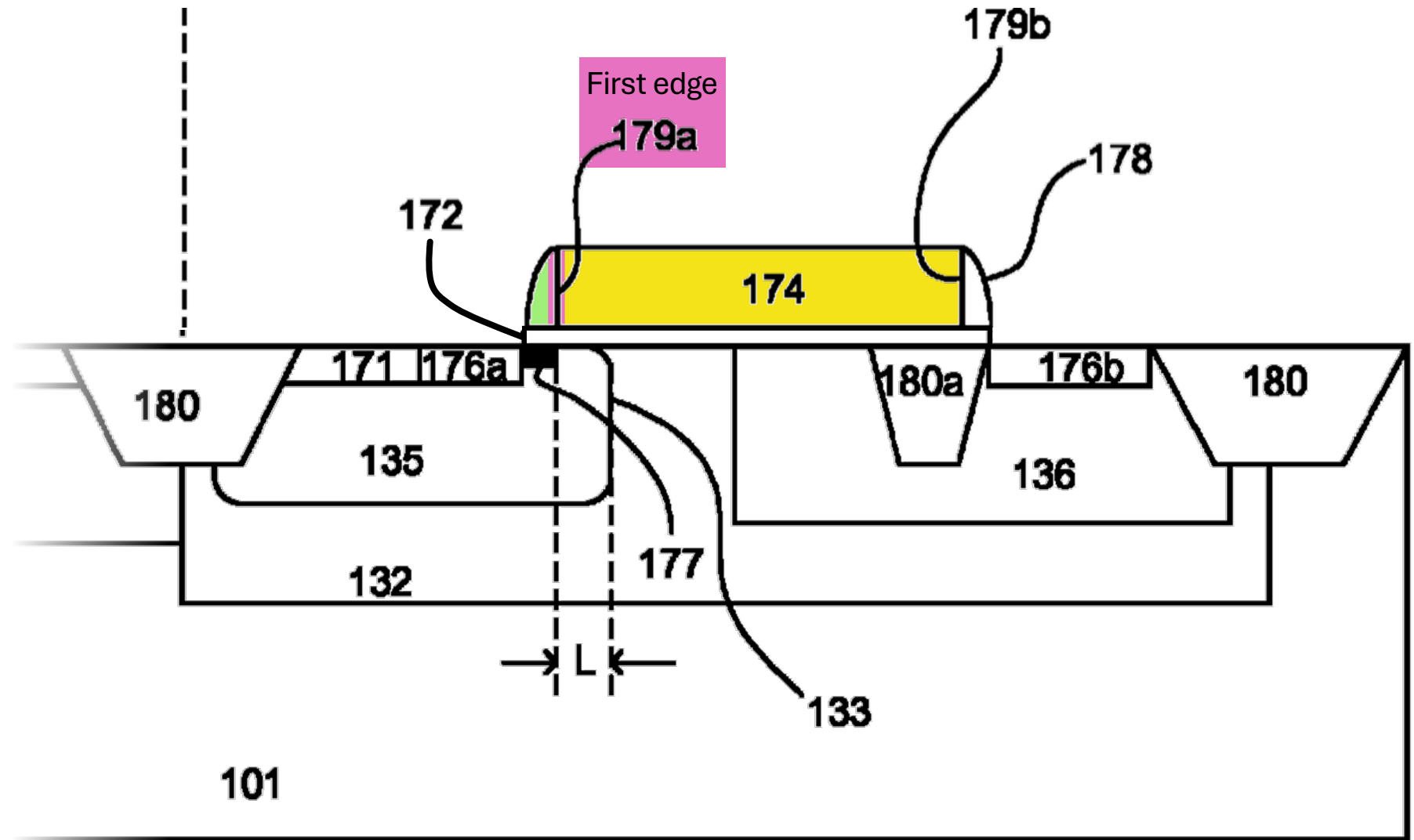
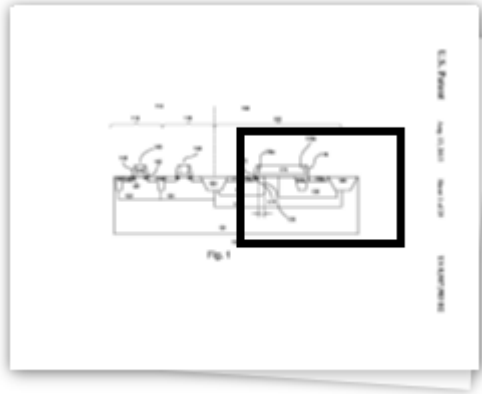


The Channel Is Where the Doped Channel Well Overlaps With the Gate Electrode



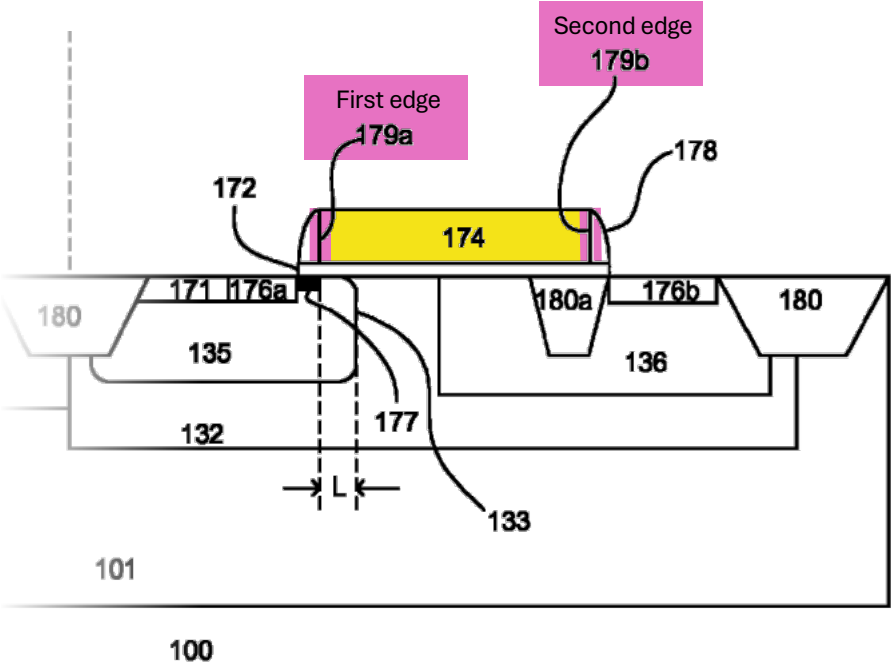
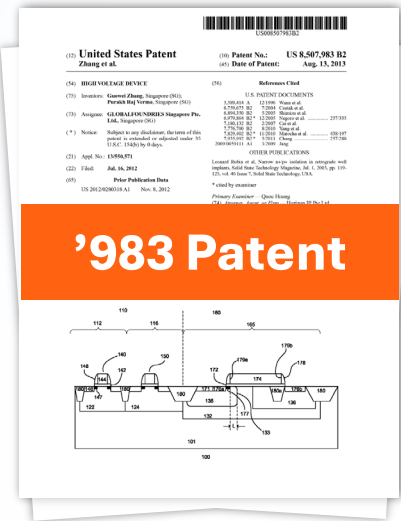
Claims: The Spacer Is “on” the First Edge of the Gate

Claim 2: “The device of claim 1 further comprising spacer on the first edge of the gate...”

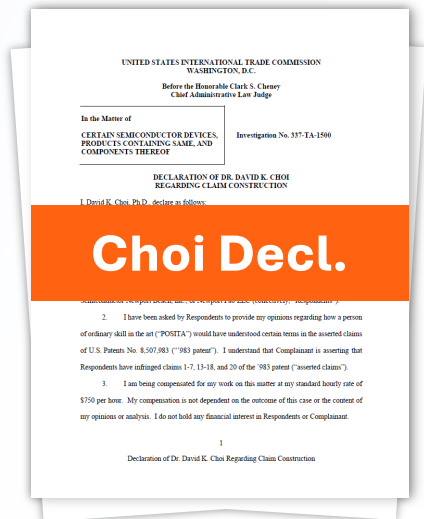


Specification: Edges of the Gate Are Those of the Electrode Over the Dielectric

As shown, a first type HV device is provided in the first type HV active region. The first type HV device, in one embodiment, comprises a LD device, such as a LDMOS device. The HV device comprises a gate with first and second edges 179a-b or sidewalls.



Dr. Choi: A POSITA Would Have Understood a Gate as an Electrode Over a Dielectric



56. Further, it is my view that, in the context of a MOSFET device, a POSITA would have understood a “gate” to be a structure that exists only where there is an electrode over a dielectric. A gate is a well-established concept in MOSFETs, where a “thin oxide layer is grown on the surface of the channel, and metal (aluminum) is deposited on it to form the gate (G). It should be clear that the name MOS is derived from the structure of the device.” (Ex. 31 at 303 (TOWER_1500_00040897)). The figure below shows the “Gate (G),” gate “Oxide,” and the gate electrode (depicted as a dark line) overlying the gate oxide.

GF's Expert: The Edges of the Gate Are the Edges of the Gate Electrode

51. For example, claim 1 recites a “gate” with a “first edge” and a “second edge.” The term “adjacent” is used to specify that a “doped channel well” and a “heavily doped region” are “adjacent to a first edge of the gate” while a “doped drift well” is “adjacent to a second edge of the gate.” ’983 patent, cl. 1. The specification similarly and consistently describes a “doped channel well” that is “adjacent to a first edge of the gate” and a “doped drift well” that is “adjacent to a second edge of the gate.” ’983 patent, 1:45-55, 1:58-66, 2:4-13. And Figure 1 confirms that spatial relationship, with a channel well 135 (yellow) in a source region next to or near a first edge of the gate 179a (red) and a drift well 136 (blue) in a drain region next to or near the second edge of the gate 179b (green):

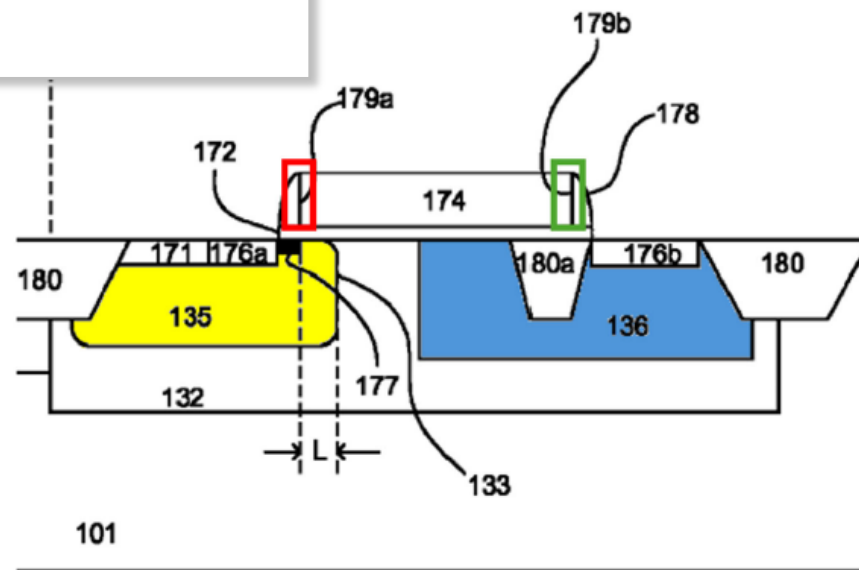
UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON DC
Before the Honorable Clark S. Cheney
Chief Administrative Law Judge

In the Matter of
CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF

Investigation No. 337-TA-1590

DECLARATION OF DAVID LIU, PH.D. IN SUPPORT OF COMPLAINANT
GLOBALFOUNDRIES' OPENING CLAIM CONSTRUCTION BRIEF

Liu Decl.



CERTIFICATE OF SERVICE

I hereby certify that on September 17, 2026, copies of the foregoing were filed with and served upon the following as indicated:

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Secretary to the Commission
U.S. INTERNATIONAL TRADE COMMISSION
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Washington, DC 20436

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